

Master's Thesis

Practical Regularized Quasi-Newton  
Methods with Inexact Function Values  
(不正確な関数値下における  
実用的な正則化準ニュートン法)

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## Abstract

Many practical optimization problems involve objective function values that are corrupted by unavoidable numerical errors. In smooth unconstrained optimization, quasi-Newton methods combined with line search are widely used due to their efficiency and scalability. These methods implicitly assume accurate function evaluations and thus they may fail to converge in noisy settings. Developing fast and robust quasi-Newton methods for such scenarios is crucial.

To address the issue, we propose a noise-tolerant regularized quasi-Newton method with an relaxed Armijo-type line search. The regularization parameter is updated using an Objective-Function-Free Optimization inspired rule that is related to the AdaGrad-Norm method. Under an error model of the objective function and standard assumptions, we establish a global convergence rate of  $\mathcal{O}(1/\varepsilon^2)$  for reaching a first-order stationary point, matching the standard convergence rate for first-order methods. We performed extensive experiments on the CUTEst benchmark collection with artificially noisy objective function evaluations, as well as with low-precision floating-point arithmetic (64-, 32-, and 16-bit). The results show that the proposed method is substantially more robust than several existing methods, while maintaining competitive practical convergence speed and computational cost.

**Keywords** Regularized Quasi-Newton Method, Numerical Stability, Numerical Error, AdaGrad-Norm, Objective-Function-Free Optimization



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# Chapter 1

## Introduction

### 1.1 Introduction

We consider the unconstrained optimization problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x), \tag{1.1}$$

where  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  is a continuously differentiable nonconvex function, and  $x \in \mathbb{R}^n$  is the optimization variable. We focus on settings where objective function evaluations are contaminated by bounded and non-diminishing numerical noise [46]. Specifically, only inexact function values  $\bar{f}(x)$  are available for optimization. Such inaccuracies are unavoidable in many realistic scenarios, including finite-precision arithmetic, simulation-based evaluations, and stochastic approximations [9, 36, 45]. When function values are contaminated by numerical noise, standard optimization algorithms may lose their theoretical guarantees, behave erratically, or terminate prematurely with unreliable solutions.

In the absence of noise and with accurate function evaluations, quasi-Newton methods, in particular the Broyden–Fletcher–Goldfarb–Shanno (BFGS) method and its Limited-memory variant (L-BFGS), are among the most efficient and widely used algorithms to solve the problem (1.1) [31, 38]. Their success is largely attributed to fast local convergence and scalability. Standard implementations [39, 49] rely on line search procedures that enforce the Wolfe conditions to guarantee sufficient descent and the positive definiteness of the Hessian approximation.

In the presence of numerical noise, these line search conditions may become unreliable. Differences in function values or directional derivatives can be dominated by rounding errors, leading to unstable step sizes, ill-conditioned Hessian approximations, or abnormal termination [40, 45]. From a computational perspective, improving the robustness of quasi-Newton methods in noisy environments is therefore crucial. In particular, it is desirable to design algorithms that (i) retain computational efficiency when function evaluations are accurate, (ii) remain stable when function values are unreliable, and (iii) guarantee convergence to first-order stationary points under mild assumptions.

In this work, we propose a noise-tolerant regularized quasi-Newton method that addresses these challenges. Our approach combines Armijo-type line search with quadratic regularization [27, 48] and Objective-Function-Free Optimization (OFFO) inspired strategy [22, 23]. When a sufficient decrease in function value is observed, the method behaves similarly to standard line search based quasi-Newton methods, preserving its practical efficiency. When numerical noise dominates, we introduce regularization on the approximate Hessian to ensure numerical stability, which yields convergence properties similar to those of OFFO methods, particularly AdaGrad-Norm [51]. The resulting algorithm is particularly well-suited for low-precision or noisy environments.

From a theoretical standpoint, we establish that the proposed method achieves the standard global convergence rate of  $\mathcal{O}(1/\varepsilon^2)$  for smooth nonconvex optimization problems, even when objective function values are corrupted by noise. Extensive numerical experiments on the CUTest

benchmark collection, across various floating-point precisions and artificial noise settings, demonstrate that the proposed method is significantly more stable than standard line search based methods in settings with significant numerical noise, while maintaining practical convergence speed.

## 1.2 Organization of This Thesis

The remainder of this thesis is organized as follows.

Chapter 2 reviews the foundations of quasi-Newton methods. In section 2.1, we briefly explain what quasi-Newton methods are. In section 2.2, we explain the modified secant condition, which is used in our numerical experiments. Since both topics are standard and well documented in the literature, we present them as a separate chapter.

Chapter 3 presents the proposed regularized quasi-Newton method along with its theoretical analysis. In section 3.1, we introduce the problem setting and review relevant background. In section 3.2, we present the proposed algorithm. In section 3.3, we establish a global convergence property of the proposed algorithm. In section 3.4, we discuss practical choices of algorithmic variables and parameters. In section 3.5, we report numerical results on the CUTEst benchmark collection. Finally, in section 3.6, we conclude the thesis with a summary and future research directions.

Our implementation is publicly available on GitHub<sup>\*1</sup>.

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<sup>\*1</sup> <https://github.com/HirokiHamaguchi/qnlab>



## Chapter 2

# Foundations of Quasi-Newton Methods

In this chapter, we review the foundations of quasi-Newton methods.

### 2.1 Quasi-Newton Methods

In this section, we discuss quasi-Newton methods. Quasi-Newton methods are based on Newton's method but aim to reduce its primary drawback, namely the high computational cost of evaluating the Hessian matrix. Instead of using the true Hessian matrix  $\nabla^2 f(x_k)$ , an approximation matrix  $B_k$  is employed, thereby achieving fast convergence while keeping the computational cost low.

Figure 2.1 shows a comparison of Newton's method, quasi-Newton methods, and gradient descent. Gradient descent updates the iterate in the direction opposite to the gradient. Although its per-iteration computational cost is the lowest, its convergence is slow. Newton's method requires the fewest iterations, but it has the highest computational cost per step. Quasi-Newton methods lie between these two extremes and strike a balance between them. In particular, when the methods are compared in terms of total computation time rather than the number of iterations, quasi-Newton methods often exhibit the best performance.

Line search based quasi-Newton methods generate a sequence  $\{x_k\}_k$  that converges to the optimal solution  $x^*$  as follows:

$$x_{k+1} = x_k - \alpha_k B_k^{-1} \nabla f(x_k) = x_k - \alpha_k H_k \nabla f(x_k).$$

Here,  $\alpha_k > 0$  is the step size determined by line search,  $B_k$  is an approximation of the Hessian matrix  $\nabla^2 f(x_k)$  at the point  $x_k$ , and  $H_k := B_k^{-1}$  denotes its inverse. This update rule closely resembles that of Newton's method, which is why it is called a quasi-Newton method.

The core of quasi-Newton methods lies in how to update  $B_k$  at each iteration so that it approaches the true Hessian matrix. Figure 2.2 illustrates this concept. First, around the current point  $x_k$ , a quadratic approximation model of the objective function  $f$  is constructed using  $B_k$ . Next, this quadratic model is minimized to obtain the next point  $x_{k+1}$ . After obtaining  $x_{k+1}$ , the approximation matrix  $B_k$  is updated to  $B_{k+1}$  using the gradient information at  $x_k$  and  $x_{k+1}$ . This procedure is repeated until convergence, which constitutes the quasi-Newton method.

#### 2.1.1 Secant Condition

Let  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  be of class  $C^2$ . Suppose that a symmetric matrix  $B_k$  is given as an approximation of the Hessian matrix  $\nabla^2 f(x_k)$ . We consider updating the approximate Hessian to  $B_{k+1}$  for the next point  $x_{k+1}$ . Although there are infinitely many possible candidates for such a matrix, it is natural to impose symmetry on  $B_{k+1}$ , since the true Hessian matrix is symmetric. For the current point  $x_k$  and the next point  $x_{k+1}$ , we define the step and the gradient difference as follows:

$$s_k := x_{k+1} - x_k, \quad y_k := \nabla f(x_{k+1}) - \nabla f(x_k).$$

## Comparison of Optimization Methods on Rosenbrock Function

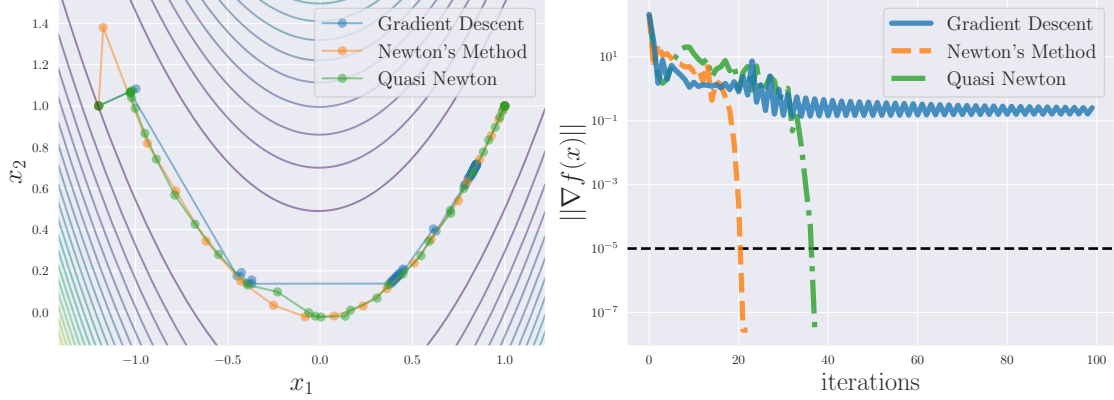


Fig. 2.1: Comparison of Newton's method, quasi-Newton methods, and gradient descent.

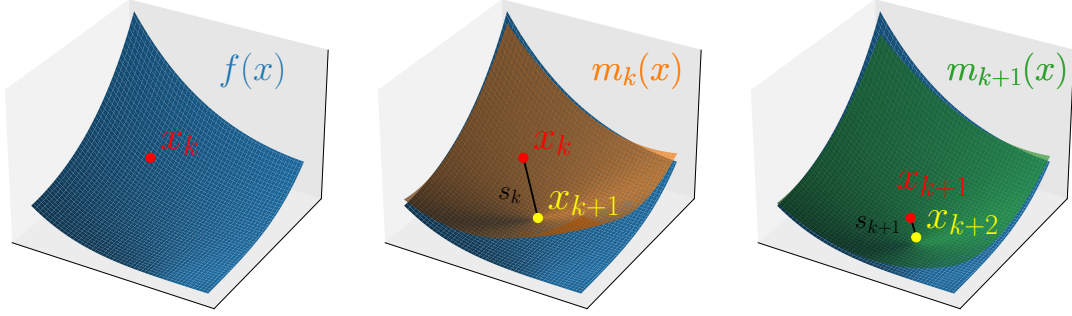


Fig. 2.2: Conceptual illustration of quasi-Newton methods. (1) Given the objective function  $f$  (blue surface) and the current point  $x_k$  (red point). (2) Move to the minimizer  $x_{k+1}$  (yellow cross) of the quadratic model  $m_k(x)$  (orange surface) constructed using the current approximate Hessian. (3) Update the quadratic model (green surface) and repeat this procedure.

In the following, we generally assume that  $s_k \neq 0$  and  $y_k^\top s_k \neq 0$ . Note that the  $s$  in  $s_k$  stands for “step.” Since the new approximation is constructed around  $x_{k+1}$ , we approximate  $\nabla f(x_k)$  using the Hessian matrix  $\nabla^2 f(x_{k+1})$  and its approximation  $B_{k+1}$  as follows:

$$\begin{aligned}\nabla f(x_k) &\approx \nabla f(x_{k+1}) + \nabla^2 f(x_{k+1})(x_k - x_{k+1}) \\ &\approx \nabla f(x_{k+1}) + B_{k+1}(x_k - x_{k+1}).\end{aligned}$$

Requiring the above approximation to hold as an equality yields

$$B_{k+1}(x_{k+1} - x_k) = \nabla f(x_{k+1}) - \nabla f(x_k).$$

Equivalently, this can be rewritten as

$$B_{k+1}s_k = y_k.$$

This relation is called the secant condition or the quasi-Newton equation.

More precisely, the secant condition can also be justified as follows. Consider a quadratic approximation model around  $x_{k+1}$ :

$$m_{k+1}(x) = f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}(x - x_{k+1}).$$

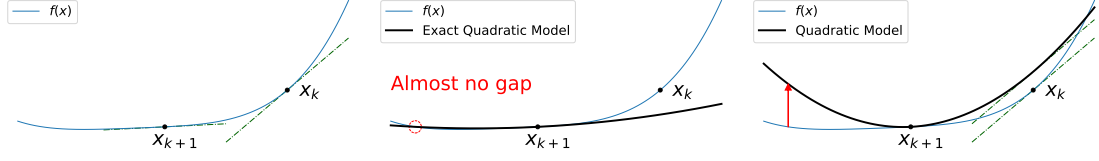


Fig. 2.3: Limitations of the standard secant condition. (a) Function values and gradients at  $x_k$  and  $x_{k+1}$  are already known. (b) The ideal quadratic model constructed from the exact Hessian fits well around the new point  $x_{k+1}$ . (c) The standard secant condition may fail to capture curvature adequately.

By construction, regardless of  $B_{k+1}$ , this model satisfies

$$\begin{cases} m_{k+1}(x_{k+1}) = f(x_{k+1}), \\ \nabla m_{k+1}(x_{k+1}) = \nabla f(x_{k+1}). \end{cases}$$

In addition, the secant condition ensures that the model satisfies

$$\begin{aligned} \nabla m_{k+1}(x_k) &= \nabla f(x_{k+1}) - B_{k+1}(x_{k+1} - x_k) && \text{(definition of the model)} \\ &= \nabla f(x_{k+1}) - (\nabla f(x_{k+1}) - \nabla f(x_k)) && \text{(secant condition)} \\ &= \nabla f(x_k). \end{aligned}$$

This implies that the secant condition ensures that the quadratic model  $m_{k+1}(x)$  accurately reflects the known information at both  $x_{k+1}$  and  $x_k$ , except for the previous function value  $f(x_k)$ .

## 2.2 Modified Secant Condition

This section explains the modified secant condition, a modification of the standard secant condition. The modified secant condition incorporates function value information in addition to gradient information, leading to more accurate Hessian approximations.

### 2.2.1 Derivation of the Modified Secant Condition

Although the secant condition is widely used in many quasi-Newton methods, it can fail to capture the curvature of the objective function accurately because it only utilizes gradient information. We illustrate this limitation in Figure 2.3. In Figure 2.3, we have two points  $x_k$  and  $x_{k+1}$  where function values and gradients are already known. We see that the ideal quadratic model constructed from the exact Hessian fits well around the new point  $x_{k+1}$ , leading to a better convergence behavior. However, the standard secant condition, which only uses gradient information, ignores function value information and may misestimate curvature significantly, resulting in a poor approximation of the true objective function.

We can overcome this limitation by utilizing the function values. The basic idea is illustrated in Figure 2.4. Even when the gradients are the same at two points  $x_k$  and  $x_{k+1}$ , natural interpolation of function values differs depending on the function values  $f(x_k)$  and  $f(x_{k+1})$ . This observation motivates the modified secant condition, which incorporates function value information to improve Hessian approximation.

In the following, we present two well-known modified secant conditions.

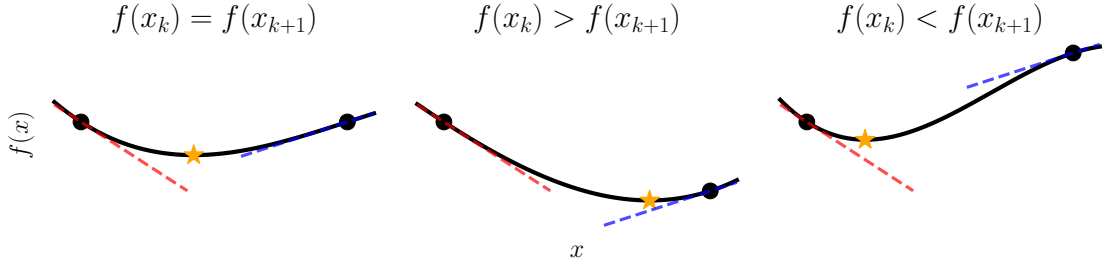


Fig. 2.4: Basic idea of the modified secant condition. Even when the gradients are the same at two points  $x_k$  and  $x_{k+1}$ , natural interpolation of function values differs depending on the function values  $f(x_k)$  and  $f(x_{k+1})$ .

#### Function-Value-Matching Modified Secant Condition

The first modification enforces the quadratic model to match the function value at the previous point [1, 52, 56]. Let us consider an alternate model with a different approximate Hessian  $B_{k+1}^F$ :

$$m_{k+1}^F(x) := f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}^F (x - x_{k+1}).$$

We require this model to satisfy

$$m_{k+1}^F(x_k) = f(x_k),$$

which enforces that the function value at the previous point is correctly modeled.

Let us derive the corresponding modified secant condition for  $B_{k+1}^F$ . Substituting the definition of  $m_{k+1}^F$  and using  $s_k = x_{k+1} - x_k$ , the condition becomes

$$\begin{aligned} f(x_k) &= f(x_{k+1}) + \nabla f(x_{k+1})^\top (x_k - x_{k+1}) + \frac{1}{2}(x_k - x_{k+1})^\top B_{k+1}^F (x_k - x_{k+1}) \\ &= f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2}s_k^\top B_{k+1}^F s_k. \end{aligned}$$

Now, instead of the standard secant condition, we assume a modified form with an additional term involving the identity matrix:

$$(B_{k+1}^F + \sigma_k^F I) s_k = y_k.$$

Here,  $\sigma_k^F \in \mathbb{R}$  is a scalar to be determined as follows. Substituting this equation yields

$$f(x_k) = f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2}s_k^\top y_k - \frac{\sigma_k^F}{2} \|s_k\|^2.$$

Solving for  $\sigma_k^F$  and using  $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$ , we obtain

$$\begin{aligned} \sigma_k^F &= \frac{2(f(x_{k+1}) - f(x_k)) - (2\nabla f(x_{k+1}) - y_k)^\top s_k}{\|s_k\|^2} \\ &= \frac{2(f(x_{k+1}) - f(x_k)) - (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2}. \end{aligned}$$

Therefore, the modified secant condition becomes

$$B_{k+1}^F s_k = \hat{y}_k^F := y_k + \frac{2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2} s_k.$$

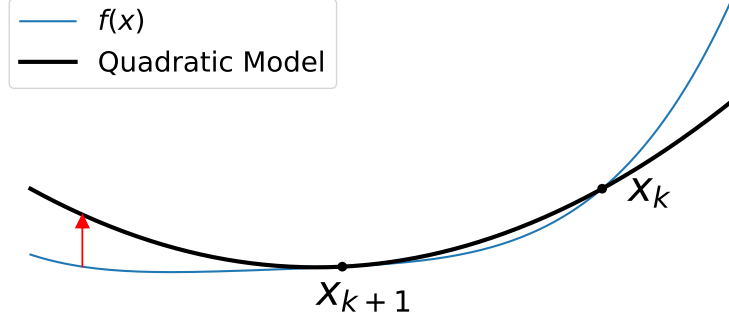


Fig. 2.5: Function-value-matching modified secant equation. The function values  $f(x_{k+1})$  and  $m_k^F(x_{k+1})$  are matched at the previous point  $x_{k+1}$ .

By using this modified  $\hat{y}_k^F$  instead of  $y_k$  in various quasi-Newton updates, we can incorporate function value information while keeping almost the same algorithm. Additionally, since condition  $s_k^\top y_k > 0$  is important for maintaining positive definiteness in BFGS update, we can consider the following modification to ensure that the inner product with  $s_k$  remains positive:

$$B_{k+1}^{F'} s_k = y_k + \frac{\max(0, 2(f(x_k) - f(x_{k+1}))) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2} s_k.$$

This is the function-value-matching modified secant condition. Refer to Figure 2.5 for an illustration.

#### Cubic-Augmented Modified Secant Condition

The second modification secant condition introduces a cubic term to the model, allowing simultaneous satisfaction of both the function value and the gradient matching at the previous point [54, 58, 59]. Let  $T_{k+1} \in \mathbb{R}^{n \times n \times n}$  be the third-order derivative tensor of  $f$  at  $x_{k+1}$  such that

$$s_k^\top (T_{k+1} s_k) s_k = \sum_{i,j,l=1}^n \partial_{x_i x_j x_l} f(x_{k+1}) s_k^{(i)} s_k^{(j)} s_k^{(l)}.$$

Here,  $\partial_{x_i x_j x_l} f$  denotes the third derivative of  $f$  with respect to  $x_i$ ,  $x_j$ , and  $x_l$ , and  $s_k^{(i)}$  is the  $i$ -th component of the vector  $s_k$ . We introduced this tensor solely for analysis purposes, and it will be eliminated from the final formula.

By incorporating this tensor term, we can define the following model:

$$\begin{aligned} m_{k+1}^C(x) := & f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2} (x - x_{k+1})^\top B_{k+1}^C (x - x_{k+1}) \\ & + \frac{1}{6} (x - x_{k+1})^\top (T_{k+1} (x - x_{k+1})) (x - x_{k+1}). \end{aligned}$$

With this model, we can enforce both function value and gradient matching at the previous point  $x_k$ . Specifically, we require

$$\begin{cases} m_{k+1}^C(x_k) = f(x_k), \\ \nabla m_{k+1}^C(x_k) = \nabla f(x_k). \end{cases}$$

By substituting the definition of  $m_{k+1}^C$  and using  $s_k = x_{k+1} - x_k$ , we can rewrite these conditions as

$$f(x_k) = f(x_{k+1}) - s_k^\top \nabla f(x_{k+1}) + \frac{1}{2} s_k^\top B_{k+1}^C s_k - \frac{1}{6} s_k^\top (T_{k+1} s_k) s_k,$$

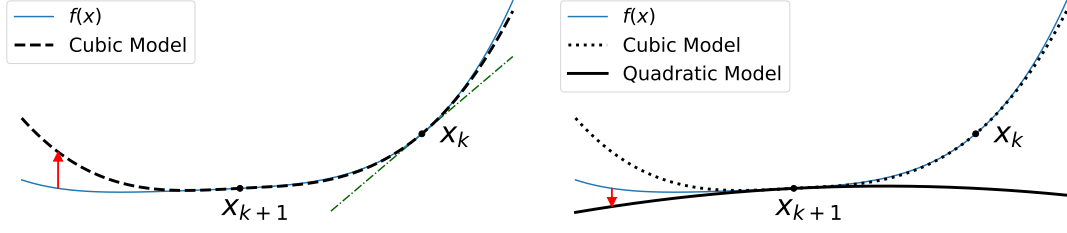


Fig. 2.6: Cubic-augmented modified secant equation. The cubic term is incorporated into the model to simultaneously match both function value and gradient at the previous point. The model is constructed using the expansion up to the quadratic term.

$$\nabla f(x_k) = \nabla f(x_{k+1}) - B_{k+1}^C s_k + \frac{1}{2}(T_{k+1} s_k) s_k.$$

Rearranging and multiplying both sides by 3 for the first equation, and taking the inner product with  $s_k$  for the second equation, we have

$$\begin{aligned} 3(f(x_k) - f(x_{k+1})) + 3s_k^\top \nabla f(x_{k+1}) &= \frac{3}{2}s_k^\top B_{k+1}^C s_k - \frac{1}{2}s_k^\top (T_{k+1} s_k) s_k, \\ -s_k^\top y_k &= -s_k^\top B_{k+1}^C s_k + \frac{1}{2}s_k^\top (T_{k+1} s_k) s_k. \end{aligned}$$

By summing up and using  $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$ , we eliminate the tensor term and obtain the scalar identity

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2}s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) + \frac{1}{2}s_k^\top y_k = \frac{1}{2}s_k^\top B_{k+1}^C s_k.$$

Again, for some scalar  $\sigma_k^C \in \mathbb{R}$ , we assume  $(B_{k+1}^C + \sigma_k^C I) s_k = y_k$ . Then, the previous equation yields

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2}s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) = -\frac{\sigma_k^C}{2} \|s_k\|^2,$$

which means

$$\sigma_k^C = -\frac{6(f(x_k) - f(x_{k+1})) + 3s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2}.$$

Thus, the modified secant condition becomes

$$B_{k+1}^C s_k = y_k + \frac{6(f(x_k) - f(x_{k+1})) + 3s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2} s_k.$$

This is a cubic-augmented modified secant condition. Refer to Figure 2.6 for an illustration.

### 2.2.2 Other Curvature Related Methods

There are several other topics related to obtaining the curvature pairs  $s_k, y_k$ . Agg-BFGS [7] is an alternative approach to managing curvature information by aggregating data rather than discarding the oldest information and adding the newest. Multi-Secant [28] extends the secant condition framework by maintaining multiple pairs of step and gradient difference vectors. In standard form we have seen, it is defined as

$$s_i = x_{i+1} - x_i, \quad y_i = \nabla f(x_{i+1}) - \nabla f(x_i), \quad (i = k-m, \dots, k)$$

In contrast, Multi-Secant centers all vectors around the latest iterate:

$$s_i = x_{k+1} - x_i, \quad y_i = \nabla f(x_{k+1}) - \nabla f(x_i). \quad (i = k-m, \dots, k)$$

This approach can lead to better approximations in some cases.

## Chapter 3

# Noise-Tolerant Regularized Quasi-Newton Method

In this chapter, we present the proposed regularized quasi-Newton method that is robust against inexact function values. This chapter is of our main contribution.

### 3.1 Preliminaries and Problem Setting

We begin by reviewing relevant background and formalizing the problem setting.

#### 3.1.1 Regularized Quasi-Newton Method

The regularized quasi-Newton method is a variant of quasi-Newton methods that does not rely heavily on line search techniques. This property is important for optimization in noisy evaluation environments. There has been renewed interest in regularized Newton-type methods, including cubic-regularization schemes [37, 42, 50], and we focus on quadratic regularization [13, 27, 30, 44, 47, 48, 57].

For  $g_k := \nabla f(x_k)$  and an approximate Hessian  $B_k \in \mathbb{R}^{n \times n}$ , regularized quasi-Newton methods compute the step as

$$x_{k+1} = x_k + d_k, \quad d_k = -(B_k + \mu_k I)^{-1} g_k,$$

with a regularization parameter  $\mu_k \geq 0$ . Combining regularization with line search is also possible and has been explored [47]. Several strategies for choosing  $\mu_k$  in nonconvex optimization has been proposed [44, 47, 48], and one of them is given by

$$\mu_k = c_1 \max(0, -\lambda_{\min}(B_k)) + c_2 \|\nabla f(x_k)\|^\delta,$$

where  $c_1 > 1$ ,  $c_2 > 0$ ,  $\delta \geq 0$ , and  $\lambda_{\min}(B_k)$  denotes the smallest eigenvalue of  $B_k$ . This choice guarantees that  $B_k + \mu_k I$  is positive definite, ensuring that  $d_k$  is a descent direction. We adapt such regularization strategies to noisy evaluation environments.

#### 3.1.2 Objective-Function-Free Optimization

Objective-Function-Free Optimization (OFFO) refers to a class of methods that avoid explicit evaluations of the objective function and rely solely on gradient information [22, 23]. Such methods are particularly attractive when function values are unreliable due to noise. Similar adaptive trust-region approaches are also discussed [20].

Prominent examples of OFFO include adaptive gradient methods such as AdaGrad [15] and



AdaGrad-Norm [51]. Their update rules are

$$\text{(AdaGrad)} \quad x_{k+1}^{(i)} = x_k^{(i)} - \frac{1}{\theta \sqrt{\varsigma + \sum_{j=0}^k (g_j^{(i)})^2}} g_k^{(i)}, \quad (i = 1, 2, \dots, n)$$

$$\text{(AdaGrad-Norm)} \quad x_{k+1} = x_k - \frac{1}{\theta \sqrt{\varsigma + \sum_{j=0}^k \|g_j\|^2}} g_k$$

respectively, where  $\varsigma > 0$  and  $\theta > 0$  are algorithmic parameters, and  $x^{(i)}$  denotes the  $i$ -th component of a vector  $x$ .

### 3.1.3 Problem Setting

We consider optimization problems in which exact objective function values are unavailable, following the recent literature on error-aware optimization [5, 10, 11, 21, 43, 46]. When both objective values and gradients are heavily corrupted by noise, reliable optimization cannot be guaranteed. Thus, meaningful optimization requires that at least one source of information be sufficiently accurate. This work focuses on problems arising from finite-precision computations or inexact evaluations, where objective function evaluations are subject to non-negligible errors while gradient information can be computed with relatively high accuracy. Settings in which objective values are accurate but gradients are noisy, such as derivative-free optimization or stochastic approximation methods [6, 45, 53], are beyond the scope of this work.

In the following, we formalize the assumptions on function value and gradient evaluations. We assume that only an error-contaminated function value  $\bar{f}(x)$  is accessible, satisfying the hybrid absolute-relative error model [25, 26]:

$$|f(x) - \bar{f}(x)| \leq \varepsilon_f \max(1, |f(x)|), \quad (3.1)$$

where  $0 \leq \varepsilon_f < 1$  is an error rate. The error is predominantly relative when  $|f(x)|$  is large and effectively absolute when  $|f(x)|$  is small, capturing the typical behavior of rounding errors in IEEE 754 floating-point arithmetic [25]. In the numerical experiments,  $\varepsilon_f$  is taken as a large multiple of the machine epsilon to account for accumulated round-off effects. Closely related models are also employed in practical stopping criteria [49].

We next specify the assumptions of gradient evaluations. In practice, termination is typically based on a gradient tolerance  $\varepsilon_{\text{gtol}}$ , and the condition  $\|\nabla \bar{f}(x)\| \leq \varepsilon_{\text{gtol}}$  is used as a stopping criterion. Hence, prior to termination, it holds that  $\|\nabla \bar{f}(x)\| > \varepsilon_{\text{gtol}}$ . If the gradient noise  $\|\nabla f(x) - \nabla \bar{f}(x)\|$  were comparable to or larger than  $\varepsilon_{\text{gtol}}$ , the stopping criterion could not be satisfied reliably, even when  $\nabla f(x) = 0$ . To ensure meaningful termination, the gradient noise must be sufficiently small relative to  $\varepsilon_{\text{gtol}}$ . Specifically, during the optimization process, we assume

$$\|\nabla f(x) - \nabla \bar{f}(x)\| \ll \varepsilon_{\text{gtol}} < \|\nabla \bar{f}(x)\|.$$

Under this regime, the computed gradient  $\nabla \bar{f}(x)$  is a sufficiently accurate approximation of the true gradient  $\nabla f(x)$  prior to termination. Accordingly, for the purpose of simplifying the theoretical analysis, we adopt the exact-gradient assumption

$$\nabla \bar{f}(x) = \nabla f(x). \quad (3.2)$$

We emphasize that this assumption is introduced solely as a modeling simplification, rather than as a claim about practical gradient accuracy. Notably, the proposed algorithm does not rely on Wolfe-type conditions, which typically require accurate gradient information at trial points, making it robust to gradient inaccuracies. Our numerical experiments in section 3.5 include cases where gradients are contaminated by noise, and they also demonstrate that the proposed method remains effective even when eq. (3.2) is violated.

## 3.2 Proposed Algorithm

We propose a new algorithm for the problem setting stated in section 3.1.3. We draw inspiration from the OFFO framework to design robust regularization and scaling strategies. Unlike purely objective-function-free methods, our algorithm exploits function values when they are numerically reliable and smoothly switches to gradient-based control otherwise. This hybrid behavior enables improved stability without sacrificing efficiency in practical optimization tasks. The pseudo-code of our algorithm is given in Algorithm 1. The minimum of the empty set is defined as  $+\infty$  in line 3. Algorithm 2 is an auxiliary line search algorithm called in Algorithm 1. In the following, we explain the three main components of Algorithm 1.

---

**Algorithm 1:** Noise Tolerant Regularized Quasi-Newton Method
 

---

**Input:**  $x_0, 0 < k_{\max}, 0 \leq \varepsilon_f < 1, 0 < \theta_{\min} \leq \theta_{\max}$  and  $0 < \varsigma < 1$

```

1  $k \leftarrow 0, g_0 \leftarrow \nabla \bar{f}(x_0), K^0 \leftarrow \emptyset, K^+ \leftarrow \emptyset$ 
2 for  $k = 0, 1, 2, \dots, k_{\max} - 1$  do
3   if  $\min_{j \in K^0} (\bar{f}(x_j) - \Delta_j) \geq \bar{f}(x_k)$  then
4      $K^0 \leftarrow K^0 \cup \{k\}$ 
5      $\mu_k \leftarrow 0$ 
6   else
7      $K^+ \leftarrow K^+ \cup \{k\}$ 
8      $\mu_k \leftarrow \theta_k \sqrt{\varsigma + \sum_{j \in K^+} \|g_j\|^2}$  with  $\theta_k \in [\theta_{\min}, \theta_{\max}]$  (section 3.4.3).
9      $d_k \leftarrow -(B_k + \mu_k I)^{-1} g_k$  with  $B_k$  (section 3.4.1).
10    Find  $\alpha_k$  and  $\Delta_k$  satisfying eq. (3.3) by Algorithm 2.
11     $x_{k+1} \leftarrow x_k + \alpha_k d_k, g_{k+1} \leftarrow \nabla \bar{f}(x_{k+1})$ 
12 end
```

---



---

**Algorithm 2:** Backtracking Line Search with Relaxed Armijo Condition
 

---

**Input:**  $x_k \in \mathbb{R}^n, d_k \in \mathbb{R}^n, g_k \in \mathbb{R}^n, 0 < c < 1$  and  $0 < \beta_{\min} < \beta_{\max} < 1$

```

1  $\alpha_k \leftarrow 1$ 
2  $\Delta_k \leftarrow \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k))$ 
3 while  $\bar{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k < \bar{f}(x_k + \alpha_k d_k)$  do
4    $\alpha_k \leftarrow \beta \alpha_k$  with  $\beta \in [\beta_{\min}, \beta_{\max}]$  (section 3.4.2).
5    $\Delta_k \leftarrow \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k))$ 
6 end
7 return  $\alpha_k$ 
```

---

Computation of the Regularized Quasi-Newton Direction (line 9): The first component is the computation of the regularized quasi-Newton direction  $d_k$  stated in section 3.1.1. Given  $g_k = \nabla \bar{f}(x_k) = \nabla f(x_k)$ , an approximate Hessian  $B_k$ , and a regularization parameter  $\mu_k$ , we compute the search direction  $d_k = -(B_k + \mu_k I)^{-1} g_k$ . A practical choice of  $B_k$  is discussed in section 3.4.1.

Relaxed Armijo Condition with an Error-Absorbing Term (line 10): The second component is the computation of the step size  $\alpha_k$  using a relaxed Armijo condition, which is defined as follows:

$$\begin{cases} \bar{f}(x_k) + c\alpha_k g_k^\top d_k + \Delta_k \geq \bar{f}(x_k + \alpha_k d_k), \\ \Delta_k = \frac{2\varepsilon_f}{1 - \varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k + \alpha_k d_k)), \end{cases} \quad (3.3)$$

where  $\Delta_k$  is the error-absorbing term that is recomputed for each  $\alpha_k$ . This relaxation guarantees the existence of  $\alpha_k \in (0, 1]$  even in the presence of numerical errors in  $\bar{f}$ . When  $d_k$  is a descent direction ( $g_k^\top d_k < 0$ ), eq. (3.3) ensures that a temporary increase of  $\bar{f}$  is at most  $\Delta_k$ . Since  $\Delta_k$  depends on  $-\bar{f}(x_k + \alpha_k d_k)$  but not on  $\bar{f}(x_k + \alpha_k d_k)$ ,  $\Delta_k$  does not increase when  $\bar{f}(x_k + \alpha_k d_k)$  becomes large. This prevents unbounded growth of  $\Delta_k$ .

Adaptive Update of the Regularization Parameter (lines 3–8): The third component is the adaptive update of the regularization parameter  $\mu_k$ . A larger  $\mu_k$  yields a more conservative step, while a smaller  $\mu_k$  allows a more aggressive quasi-Newton step, especially when  $\mu_k = 0$ . In practical settings, setting  $\mu_k = 0$  is preferred, since this allows us to fully exploit the quasi-Newton approximation and typically leads to practically faster convergence. Let us partition the iteration indices  $K := \{0, 1, 2, \dots, k_{\max} - 1\}$  into two sets:

$$K^0 := \{k \in K \mid \mu_k = 0\}, \quad K^+ := \{k \in K \mid \mu_k > 0\}. \quad (3.4)$$

We set  $\mu_k = 0$  if and only if the following condition is satisfied:

$$\min_{j \in K^0, j < k} (\bar{f}(x_j) - \Delta_j) \geq \bar{f}(x_k). \quad (3.5)$$

Note that  $k = 0$  satisfies eq. (3.5) since  $K^0$  is empty. This condition restricts  $\mu_k = 0$  to iterations for which a sufficient decrease in  $\bar{f}$  has been observed. Thereby, while  $\bar{f}$  can temporarily increase, this condition theoretically ensures that  $\bar{f}$  does not grow unboundedly or oscillate indefinitely.

Otherwise, to ensure convergence to first-order stationary points, we apply an OFFO-based update

$$\mu_k = \theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}, \quad (3.6)$$

as described in section 3.1.2. Since OFFO schemes do not rely on function values, this update is robust to numerical errors in  $\bar{f}$ .

### 3.3 Theoretical Analysis

In this section, we establish the global convergence properties of Algorithm 1 under the problem setting in section 3.1.3, namely, eqs. (3.1) and (3.2).

#### 3.3.1 Assumptions

Throughout this section, we make the following three assumptions. The first assumption ensures that the objective function is bounded below.

**Assumption 1.** *The function  $f$  is bounded below, meaning that for all  $x \in \mathbb{R}^n$ ,*

$$f(x) \geq f^* := \inf_{x \in \mathbb{R}^n} f(x) > -\infty.$$

By the triangle inequality and eq. (3.1), we have  $\bar{f}(x) \geq f(x) - \varepsilon_f \max(1, |f(x)|)$ . Since the function  $g(t) = t - \varepsilon_f \max(1, |t|)$  is non-decreasing for  $0 \leq \varepsilon_f < 1$ , it follows that  $\inf_x g(f(x)) = g(f^*)$ . Consequently, we can also lower bound  $\bar{f}$  as

$$\bar{f}(x) \geq \bar{f}^* := \inf_{x \in \mathbb{R}^n} (f(x) - \varepsilon_f \max(1, |f(x)|)) = f^* - \varepsilon_f \max(1, |f^*|) > -\infty. \quad (3.7)$$

The second assumption ensures smoothness of the objective function.

**Assumption 2.** The function  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  is  $L$ -smooth, meaning that there exists a constant  $L > 0$  such that for all  $x, y \in \mathbb{R}^n$ ,

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2.$$

The third assumption requires the matrices  $\{B_k\}_{k=0}^{k_{\max}-1}$  to be uniformly bounded and uniformly positive definite.

**Assumption 3.** There exist constants  $0 < m \leq M$  such that all  $B_k$  satisfy

$$mI \preceq B_k \preceq MI.$$

As we will discuss in section 3.4, the standard BFGS update with a slight modification guarantees Assumption 3 in practice. By Assumption 3, we can derive useful bounds involving  $g_k$  and  $d_k$ . Using  $d_k = -(B_k + \mu_k I)^{-1} g_k$  (line 9) and Assumption 3 asserting  $(m + \mu_k)I \preceq B_k + \mu_k I \preceq (M + \mu_k)I$ , we obtain the following bounds:

$$-g_k^\top d_k = d_k^\top (B_k + \mu_k I) d_k \geq (m + \mu_k) \|d_k\|^2 \geq m \|d_k\|^2, \quad (3.8)$$

$$-g_k^\top d_k = g_k^\top (B_k + \mu_k I)^{-1} g_k \geq \frac{1}{M + \mu_k} \|g_k\|^2, \quad (3.9)$$

$$\|d_k\|^2 = \|(B_k + \mu_k I)^{-1} g_k\|^2 \leq \frac{1}{(m + \mu_k)^2} \|g_k\|^2. \quad (3.10)$$

### 3.3.2 Error Bound on Function Evaluations

We first establish useful properties of the error-absorbing term  $\Delta_k$  defined in eq. (3.3). In the following, we define the actual and observed function decreases as

$$\delta_k := f(x_k) - f(x_{k+1}), \quad (3.11)$$

$$\bar{\delta}_k := \bar{f}(x_k) - \bar{f}(x_{k+1}). \quad (3.12)$$

The first property states that the error between the actual values difference  $\delta_k$  and the observed values difference  $\bar{\delta}_k$  can be bounded by  $\Delta_k$ .

**Lemma 1.** If  $x_k$  and  $x_{k+1}$  satisfy  $f(x_{k+1}) \leq f(x_k)$ , it holds that

$$\delta_k \leq \bar{\delta}_k + \Delta_k. \quad (3.13)$$

*Proof.* We start from general observations. We first claim that, for all  $x \in \mathbb{R}^n$ ,

$$\begin{cases} \max(1, +f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, +\bar{f}(x)), \\ \max(1, -f(x)) \leq \frac{1}{1-\varepsilon_f} \max(1, -\bar{f}(x)). \end{cases} \quad (3.14)$$

We prove the case with the plus sign. If  $+f(x) \leq 1$ , then  $\max(1, +f(x)) = 1 \leq \frac{1}{1-\varepsilon_f}$  and the inequality is trivial. If  $+f(x) > 1$ , eq. (3.1) implies

$$f(x) - \bar{f}(x) \leq |f(x) - \bar{f}(x)| \leq \varepsilon_f \max(1, |f(x)|) = +\varepsilon_f f(x). \quad (3.15)$$

Rearranging this yields

$$\max(1, +f(x)) = +f(x) \leq +\frac{1}{1-\varepsilon_f} \bar{f}(x) \leq \frac{1}{1-\varepsilon_f} \max(1, +\bar{f}(x)).$$

The negative case can be treated analogously, and this completes the proof of eq. (3.14).

We also have the following general bound for all  $k \in K$ :

$$\begin{aligned} |\delta_k - \bar{\delta}_k| &= |(f(x_k) - \bar{f}(x_k)) - (f(x_{k+1}) - \bar{f}(x_{k+1}))| && \text{(rearrangement)} \\ &\leq |f(x_k) - \bar{f}(x_k)| + |f(x_{k+1}) - \bar{f}(x_{k+1})| && \text{(triangle inequality)} \\ &\leq \varepsilon_f \max(1, |f(x_k)|) + \varepsilon_f \max(1, |f(x_{k+1})|) && \text{(definition in eq. (3.1))} \\ &\leq 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|). && \text{(max property)} \end{aligned} \quad (3.16)$$

With these preparations, we now prove the lemma. From the condition  $f(x_{k+1}) \leq f(x_k)$ , eq. (3.16) can be further bounded as

$$\begin{aligned} 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|) &= 2\varepsilon_f \max(1, f(x_k), -f(x_{k+1})) && (f(x_{k+1}) \leq f(x_k)) \\ &\leq \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_{k+1})) && \text{(eq. (3.14))} \\ &= \Delta_k, && \text{(eq. (3.3))} \end{aligned}$$

which yields the desired inequality with  $\delta_k - \bar{\delta}_k \leq |\delta_k - \bar{\delta}_k| \leq \Delta_k$ .  $\square$

We also derive the following lemma for later use.

**Lemma 2.** *Under Assumption 3, it holds that*

$$\bar{\delta}_k \leq \delta_k + \frac{1+\varepsilon_f}{1-\varepsilon_f} \Delta_k.$$

*Proof.* Since  $-g_k^\top d_k \geq 0$  from Assumption 3 and eq. (3.8), eq. (3.3) implies

$$\bar{f}(x_k) - \bar{f}(x_{k+1}) \geq -c\alpha_k g_k^\top d_k - \Delta_k \geq -\Delta_k. \quad (3.17)$$

Thus, using the general result in eq. (3.16), we can further bound as

$$\begin{aligned} |\delta_k - \bar{\delta}_k| &\leq 2\varepsilon_f \max(1, |f(x_k)|, |f(x_{k+1})|) && \text{(eq. (3.16))} \\ &\leq \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k), -\bar{f}(x_k), \bar{f}(x_{k+1}), -\bar{f}(x_{k+1})) && \text{(eq. (3.14))} \\ &\leq \frac{2\varepsilon_f}{1-\varepsilon_f} \max(1, \bar{f}(x_k) + \Delta_k, -\bar{f}(x_{k+1}) + \Delta_k) && \text{(eq. (3.17))} \\ &\leq \frac{2\varepsilon_f}{1-\varepsilon_f} (\max(1, \bar{f}(x_k), -\bar{f}(x_{k+1})) + \Delta_k) \\ &= \left(1 + \frac{2\varepsilon_f}{1-\varepsilon_f}\right) \Delta_k = \frac{1+\varepsilon_f}{1-\varepsilon_f} \Delta_k, && \text{(eq. (3.3))} \end{aligned}$$

which yields the desired inequality with  $\bar{\delta}_k - \delta_k \leq |\delta_k - \bar{\delta}_k| \leq \frac{1+\varepsilon_f}{1-\varepsilon_f} \Delta_k$ .  $\square$

### 3.3.3 Backtracking Stage

We next analyze the line search in Algorithm 2. Recall that the parameter  $c$  in Algorithm 2 satisfy  $0 < c < 1$  and  $0 < m$  from Assumption 3.

**Lemma 3.** *Under Assumptions 2 and 3, the backtracking procedure in Algorithm 2 terminates whenever the step size  $\alpha_k$  satisfies*

$$\alpha_k \leq \frac{2(1-c)m}{L}. \quad (3.18)$$

*Proof.* Under the  $L$ -smoothness of  $f$  (Assumption 2), we always have

$$\begin{aligned} f(x_k) - f(x_k + \alpha_k d_k) &\geq -\alpha_k g_k^\top d_k - \frac{L}{2} \alpha_k^2 \|d_k\|^2 \\ &= -c\alpha_k g_k^\top d_k + \alpha_k \left( -(1-c)g_k^\top d_k - \frac{L\alpha_k}{2} \|d_k\|^2 \right). \end{aligned} \quad (3.19)$$

From Assumption 3 and eq. (3.8), we have  $-g_k^\top d_k \geq m\|d_k\|^2$ . Under eq. (3.18), we have

$$-(1-c)g_k^\top d_k - \frac{L\alpha_k}{2} \|d_k\|^2 \geq \left( (1-c)m - \frac{L\alpha_k}{2} \right) \|d_k\|^2 \geq 0. \quad (3.20)$$

Combining eqs. (3.19) and (3.20) yields

$$f(x_k) - f(x_k + \alpha_k d_k) \geq -c\alpha_k g_k^\top d_k.$$

Since  $-g_k^\top d_k \geq m\|d_k\|^2 \geq 0$ , the condition of Lemma 1 is satisfied with  $x_{k+1} = x_k + \alpha_k d_k$ . Thus we obtain  $\delta_k \leq \bar{\delta}_k + \Delta_k$ , i.e.,

$$-c\alpha_k g_k^\top d_k \leq f(x_k) - f(x_k + \alpha_k d_k) \leq \bar{f}(x_k) - \bar{f}(x_k + \alpha_k d_k) + \Delta_k.$$

This inequality means that the relaxed Armijo condition (eq. (3.3)) is satisfied and thus the backtracking procedure terminates. This completes the proof.  $\square$

Now, let us analyze Algorithm 2 using Lemma 3. Define

$$\bar{\alpha} := \min \left( \beta_{\min} \frac{2(1-c)m}{L}, 1 \right) > 0. \quad (3.21)$$

The following summarizes the behavior of the backtracking line search.

**Proposition 1.** *Under Assumptions 2 and 3, the backtracking line search in Algorithm 2 terminates after finitely many rejections, and the step size  $\alpha_k$  satisfies*

$$\alpha_k \geq \bar{\alpha}.$$

*Proof.* The line 4 in Algorithm 2 multiplies  $\alpha_k$  by  $\beta$  at each rejection, satisfying  $0 < \beta_{\min} \leq \beta \leq \beta_{\max} < 1$ . For the number of backtracking rejections  $r$ , we have  $\alpha_k \leq \beta_{\max}^r$ . Thus,  $r$  is finite since the backtracking procedure terminates for sufficiently small  $\alpha_k$  by Lemma 3. If  $r = 0$ ,  $\alpha_k = 1 \geq \bar{\alpha}$ . Otherwise,  $\alpha_k \geq \beta_{\min} \alpha'_k$  where  $\alpha'_k$  is the last rejected step size. By Lemma 3, we have  $\alpha'_k > \frac{2(1-c)m}{L}$ , and thus

$$\alpha_k \geq \beta_{\min} \alpha'_k > \beta_{\min} \frac{2(1-c)m}{L} \geq \bar{\alpha}. \quad (3.22)$$

This completes the proof.  $\square$

### 3.3.4 Bound for the case $\mu_k = 0$

In this subsection, we bound the sum of actual function decreases  $\delta_k$  over  $k \in K^0$ . Recall that  $K^0$  denotes the set of iteration indices  $k$  such that  $\mu_k = 0$ , as defined in eq. (3.4).

To this end, we establish that the sum of  $\Delta_k$  over  $k \in K^0$  is bounded, where  $\Delta_k$  is the error-absorbing term in the relaxed Armijo condition (eq. (3.3)). We introduce the following constant:

$$\Delta_{\max} := \frac{2\varepsilon_f}{1 - \varepsilon_f} \max(1, \bar{f}(x_0), -\bar{f}^*). \quad (3.23)$$

**Lemma 4.** *Under Assumptions 1 to 3, the sum of  $\Delta_k$  over  $k \in K^0$  satisfies*

$$\sum_{k \in K^0} \Delta_k \leq \bar{f}(x_0) - \bar{f}^* + \Delta_{\max}.$$

*Proof.* Let us define the last index of  $|K^0|$  as  $\ell$ , and enumerate the elements of  $K^0$  in increasing order as  $K^0 = (k_i)_{i=0}^{\ell}$ . For all  $i < \ell$ , using eq. (3.5) with  $j = k_i$  and  $k = k_{i+1}$ , we obtain

$$\Delta_{k_i} \leq \bar{f}(x_{k_i}) - \bar{f}(x_{k_{i+1}}). \quad (3.24)$$

We also have  $\bar{f}(x_0) \geq \bar{f}(x_{k_\ell})$  from eq. (3.5) with  $k = k_\ell$ , or just  $x_0 = x_{k_\ell}$ . Thus, by Assumption 1, we have  $\Delta_{k_\ell} = \frac{2\varepsilon_f}{1 - \varepsilon_f} \max(1, \bar{f}(x_{k_\ell}), -\bar{f}(x_{k_{\ell+1}})) \leq \Delta_{\max}$ . Hence, we have

$$\begin{aligned} \sum_{k \in K^0} \Delta_k &\leq \sum_{i=0}^{\ell-1} (\bar{f}(x_{k_i}) - \bar{f}(x_{k_{i+1}})) + \Delta_{k_\ell} && \text{(eq. (3.24))} \\ &= \bar{f}(x_0) - \bar{f}(x_{k_\ell}) + \Delta_{k_\ell} && \text{(telescoping sum)} \\ &\leq \bar{f}(x_0) - \bar{f}^* + \Delta_{\max}. && (\bar{f}(x_{k_\ell}) \geq \bar{f}^* \text{ and } \Delta_{k_\ell} \leq \Delta_{\max}) \end{aligned}$$

This completes the proof.  $\square$

We now bound the sum of  $\delta_k = f(x_k) - f(x_{k+1})$  over  $k \in K^0$  with  $\|g_k\|^2$  as follows.

**Lemma 5.** *Under Assumptions 1 to 3, the sum of  $\delta_k$  over  $k \in K^0$  satisfies*

$$\sum_{k \in K^0} \delta_k \geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1 - \varepsilon_f} (\bar{f}(x_0) - \bar{f}^* + \Delta_{\max}).$$

*Proof.* The relaxed Armijo condition (eq. (3.3)) at iteration  $k \in K^0$  yields

$$\begin{aligned} \bar{\delta}_k + \Delta_k &= \bar{f}(x_k) - \bar{f}(x_{k+1}) + \Delta_k && \text{(eq. (3.12))} \\ &\geq -c\alpha_k g_k^\top d_k && \text{(eq. (3.3))} \\ &\geq \frac{c\bar{\alpha}}{M + \mu_k} \|g_k\|^2 && \text{(Proposition 1 and eq. (3.9))} \\ &= \frac{c\bar{\alpha}}{M} \|g_k\|^2. && (\mu_k = 0 \text{ since } k \in K^0) \end{aligned} \quad (3.25)$$

Using Lemma 2, we also have

$$\bar{\delta}_k + \Delta_k \leq \delta_k + \frac{1 + \varepsilon_f}{1 - \varepsilon_f} \Delta_k + \Delta_k = \delta_k + \frac{2}{1 - \varepsilon_f} \Delta_k. \quad (3.26)$$

Then, combining eqs. (3.25) and (3.26) and summing up over  $k \in K^0$  gives

$$\sum_{k \in K^0} \frac{c\bar{\alpha}}{M} \|g_k\|^2 \leq \sum_{k \in K^0} \delta_k + \frac{2}{1 - \varepsilon_f} \sum_{k \in K^0} \Delta_k \leq \sum_{k \in K^0} \delta_k + \frac{2}{1 - \varepsilon_f} (\bar{f}(x_0) - \bar{f}^* + \Delta_{\max}),$$

where the last inequality follows from Lemma 4. Rearranging this yields the desired inequality.  $\square$

### 3.3.5 Bound for the case $\mu_k > 0$

Next, we show that the sum of  $\delta_k$  over  $k \in K^+$  is bounded. Recall that  $K^+$  be a set of iteration indices  $k$  such that  $\mu_k > 0$  as defined in eq. (3.4), and that  $\mu_k$  is defined as  $\theta_k \sqrt{\varsigma + \sum_{j \in K^+, j < k} \|g_j\|^2}$  with  $\theta_k \in [\theta_{\min}, \theta_{\max}]$  in line 8 of Algorithm 1.

Before proceeding to the main proof, we present Lemma 6, which is standard inequalities in the AdaGrad-Norm algorithm analysis [51, Lemma 3.2], [23, Lemma 3.1]. Its proof is deferred to section A.1.

**Lemma 6.** *In Algorithm 1, we have*

$$\begin{aligned} \sum_{k \in K^+} \frac{\|g_k\|^2}{\mu_k^2} &\leq \frac{1}{\theta_{\min}^2} \left( \ln \left( \varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right), \\ \sum_{k \in K^+} \frac{\|g_k\|^2}{\mu_k} &\geq \frac{1}{\theta_{\max}} \left( \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right). \end{aligned}$$

We now bound the sum of  $\delta_k$  over  $k \in K^+$  with  $\|g_k\|^2$ . We define the following constants for later use:

$$C_1 := \frac{\bar{\alpha}}{\theta_{\max}}, \quad C_2 := \frac{M\bar{\alpha} + L/2}{\theta_{\min}^2}. \quad (3.27)$$

**Lemma 7.** *Under Assumptions 2 and 3, the sum of  $\delta_k$  over  $k \in K^+$  satisfies*

$$\sum_{k \in K^+} \delta_k \geq C_1 \left( \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right) - C_2 \left( \ln \left( \varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right).$$

*Proof.* From the  $L$ -smoothness (Assumption 2), for  $k \in K^+$  we have

$$\begin{aligned} \delta_k &= f(x_k) - f(x_k + \alpha_k d_k) && \text{(eq. (3.11) and } x_{k+1} = x_k + \alpha_k d_k) \\ &\geq -\alpha_k g_k^\top d_k - \frac{\alpha_k^2 L}{2} \|d_k\|^2 && \text{(Assumption 2)} \\ &\geq \left( \frac{\alpha_k}{M + \mu_k} - \frac{\alpha_k^2 L}{2(m + \mu_k)^2} \right) \|g_k\|^2 && \text{(eqs. (3.9) and (3.10))} \\ &\geq \left( \frac{\bar{\alpha}}{M + \mu_k} - \frac{L}{2(m + \mu_k)^2} \right) \|g_k\|^2. && \text{(Proposition 1 and } \alpha_k \leq 1) \end{aligned} \quad (3.28)$$

We can further bound the terms in eq. (3.28) as

$$\frac{\bar{\alpha}}{M + \mu_k} = \frac{\bar{\alpha}}{\mu_k} \frac{1}{1 + \frac{M}{\mu_k}} \geq \frac{\bar{\alpha}}{\mu_k} \left( 1 - \frac{M}{\mu_k} \right) = \frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha}}{\mu_k^2}, \quad \frac{L}{2(m + \mu_k)^2} \leq \frac{L}{2\mu_k^2}. \quad (3.29)$$



Thus, applying eq. (3.29) to eq. (3.28) yields

$$\delta_k \geq \left( \frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha} + L/2}{\mu_k^2} \right) \|g_k\|^2. \quad (3.30)$$

Summing up eq. (3.30) over  $k \in K^+$  and applying Lemma 6 yield

$$\begin{aligned} \sum_{k \in K^+} \delta_k &\geq \sum_{k \in K^+} \left( \frac{\bar{\alpha}}{\mu_k} - \frac{M\bar{\alpha} + L/2}{\mu_k^2} \right) \|g_k\|^2 \\ &\geq \frac{\bar{\alpha}}{\theta_{\max}} \left( \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - \sqrt{\varsigma} \right) - \frac{M\bar{\alpha} + L/2}{\theta_{\min}^2} \left( \ln \left( \varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) - \ln(\varsigma) \right), \end{aligned}$$

which yields the desired result.  $\square$

### 3.3.6 Global Convergence Result

Finally, we combine the results for  $k \in K^0$  (Lemma 5) and  $k \in K^+$  (Lemma 7) to derive a bound on the total sum of  $\|g_k\|^2$ . We define the constant  $F$  as follows:

$$F := f(x_0) - f^* + \frac{2}{1 - \varepsilon_f} \left( \bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right) + C_1 \sqrt{\varsigma} - C_2 \ln(\varsigma). \quad (3.31)$$

Then, we can obtain the following unified bound of the sum of  $\|g_k\|^2$ .

**Lemma 8.** *Under the assumptions, the sum of  $\|g_k\|^2$  over  $k \in K^0, K^+$  satisfies*

$$F \geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + C_1 \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - C_2 \ln \left( \varsigma + \sum_{k \in K^+} \|g_k\|^2 \right),$$

*Proof.* Since  $f(x_{k_{\max}}) \geq f^*$  by Assumption 1 and  $K = K^0 \cup K^+$ , we have

$$f(x_0) - f^* \geq f(x_0) - f(x_{k_{\max}}) = \sum_{k \in K} \delta_k = \sum_{k \in K^0} \delta_k + \sum_{k \in K^+} \delta_k.$$

From Lemmas 5 and 7, we can further bound as

$$\begin{aligned} f(x_0) - f^* &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 - \frac{2}{1 - \varepsilon_f} \left( \bar{f}(x_0) - \bar{f}^* + \Delta_{\max} \right) \\ &\quad + C_1 \sqrt{\varsigma + \sum_{k \in K^+} \|g_k\|^2} - C_1 \sqrt{\varsigma} - C_2 \ln \left( \varsigma + \sum_{k \in K^+} \|g_k\|^2 \right) + C_2 \ln(\varsigma). \end{aligned}$$

Substituting eq. (3.31) yields the desired bound, concluding the proof.  $\square$

Now, we establish the following global convergence result for Algorithm 1.

**Theorem 1** (Global Convergence). *Under Assumptions 1 to 3, the average of squared gradient norms generated by Algorithm 1 satisfies:*

$$\frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 = \mathcal{O} \left( \frac{1}{k_{\max}} \right).$$

*Proof.* Let us define the function  $\phi: \mathbb{R}_{>0} \rightarrow \mathbb{R}$  as

$$\phi(G) := \frac{C_1}{2} \sqrt{G} - C_2 \ln(G). \quad (3.32)$$

Since  $\phi'(G) = \frac{C_1}{4\sqrt{G}} - \frac{C_2}{G}$ , its minimum over  $G > 0$  is attained at  $G^* := \left(\frac{4C_2}{C_1}\right)^2$  with

$$\phi(G^*) = 2C_2 \left(1 - \ln\left(\frac{4C_2}{C_1}\right)\right) = 2C_2 \ln\left(\frac{eC_1}{4C_2}\right),$$

where  $e$  is the base of the natural logarithm. Thus, Lemma 8 yields

$$\begin{aligned} F &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} + \phi\left(\sum_{k \in K^+} \|g_k\|^2 + \varsigma\right) \\ &\geq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} + \phi(G^*), \end{aligned}$$

which is equivalent to

$$0 \leq \frac{c\bar{\alpha}}{M} \sum_{k \in K^0} \|g_k\|^2 + \frac{C_1}{2} \sqrt{\sum_{k \in K^+} \|g_k\|^2} \leq F' \quad (3.33)$$

where

$$\begin{aligned} F' &:= F - \phi(G^*) \\ &= f(x_0) - f^* + \frac{2}{1 - \varepsilon_f} (\bar{f}(x_0) - \bar{f}^* + \Delta_{\max}) + C_1 \sqrt{\varsigma} - C_2 \ln(\varsigma) - 2C_2 \ln\left(\frac{eC_1}{4C_2}\right). \end{aligned}$$

It remains to bound  $\sum_{k \in K} \|g_k\|^2 = \sum_{k \in K^0} \|g_k\|^2 + \sum_{k \in K^+} \|g_k\|^2$  given the constraint in eq. (3.33). More generally, let  $a, b > 0$ ,  $c \geq 0$  and  $x, y \geq 0$ , and consider the maximization problem  $x + y$  subject to  $ax + b\sqrt{y} \leq c$ . The maximum is achieved on the boundary  $ax + b\sqrt{y} = c$  since  $x + y$  is increasing in both variables. On this boundary, we can write  $y = \left(\frac{c-ax}{b}\right)^2$ , and thus  $x + y$  is convex on an interval  $0 \leq x \leq c/a$ , attaining its maximum at one of the endpoints. Evaluating at the endpoints yields the candidates  $(x, y) = (c/a, 0)$  and  $(x, y) = (0, (c/b)^2)$ , and therefore the maximum of  $x + y$  is  $\max(c/a, (c/b)^2)$ . Applying this result to eq. (3.33) gives

$$\begin{aligned} \frac{1}{k_{\max}} \sum_{k=0}^{k_{\max}-1} \|g_k\|^2 &= \frac{1}{k_{\max}} \left( \sum_{k \in K^0} \|g_k\|^2 + \sum_{k \in K^+} \|g_k\|^2 \right) \quad (K = K^0 \cup K^+) \\ &\leq \frac{1}{k_{\max}} \max\left(\frac{M}{c\bar{\alpha}} F', \frac{4}{C_1^2} F'^2\right), \end{aligned} \quad (3.34)$$

which completes the proof.  $\square$

As a remark, the function  $\phi(G)$  in eq. (3.32) is closely related to the Lambert  $W$  function [12, 23]. For  $x \in [-1/e, 0)$ , the equation  $ye^y = x$  admits two real negative solutions, and the smaller one is given by the second branch of Lambert  $W$  function  $y = W_{-1}(x) < 0$ . Using this function, the equation  $\phi(G) = 0$  can be solved in closed form, which leads to a sharper bound in Theorem 1. Since this is not essential for the main argument, we omit the details and refer to [23].

We next interpret eq. (3.34) in the proof of Theorem 1. The contribution of the indices in  $K^0$  is analogous to the classical analysis of gradient descent. Indeed, for gradient descent with fixed step size  $\alpha = 1/L$ , one has

$$\sum_{j=0}^k \|g_j\|^2 \leq 2L(f(x_0) - f^*) = \frac{2(f(x_0) - f^*)}{\alpha}.$$

Thus, the term associated with  $K^0$  reflects the standard  $\mathcal{O}(f(x_0) - f^*)$  dependence of first-order methods since  $F'$  contains the initial gaps  $f(x_0) - f^*$  and  $\bar{f}(x_0) - \bar{f}^*$ . In contrast, the contribution of  $K^+$  resembles the behavior of AdaGrad-Norm, whose convergence bound scales quadratically with the initial optimality gap [51].

Finally, it is well-known that Theorem 1 implies that the iteration complexity:

$$\min \{k \mid \|\nabla f(x_k)\| \leq \varepsilon_{\text{gtol}}\} = \mathcal{O}(1/\varepsilon_{\text{gtol}}^2).$$

This is a standard convergence guarantee for first-order optimization algorithms applied to  $L$ -smooth nonconvex functions.

### 3.4 Practical Choices of Algorithmic Variables

In sections 3.2 and 3.3, we introduced Algorithm 1 and established its convergence properties. In this section, we discuss practical choices of algorithmic variables and parameters that lead to fast and stable performance. The quantities to be discussed here are the matrices  $B_k$ , the step sizes  $\alpha_k$ , and the regularization parameters  $\mu_k$ .

#### 3.4.1 Approximate Hessian $B_k$

We first discuss the construction of the matrices  $B_k$  used in line 9 of Algorithm 1, using the L-BFGS method. The purpose of this subsection is to show that  $B_k$  can satisfy Assumption 3, the uniform spectral bound, with a slight modification and selection of curvature pairs.

Construction of  $B_k$ : We briefly note the L-BFGS method and the curvature pair notation [7, 31, 38]. For an iteration  $k$ , we define

$$s_k := x_{k+1} - x_k, \quad y_k := g_{k+1} - g_k \quad (3.35)$$

where  $g_k$  and  $g_{k+1}$  are the gradients at  $x_k$  and  $x_{k+1}$ , respectively. Starting from an initial matrix, an approximate Hessian is obtained by successively applying BFGS updates to a sequence of curvature pairs. Let  $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$  be a collection of curvature pairs, where  $k_i$  denotes the iteration index at which the  $i$ -th pair was generated. Starting from the initial matrix  $\gamma I$  with  $\gamma = \|y_{k_0}\|^2 / y_{k_0}^\top s_{k_0}$ , we define  $\text{BFGS}(\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1})$  as the matrix  $B_k$  obtained by sequentially applying the BFGS update using these pairs in the given order. Here,  $p$  denotes the maximum number of stored curvature pairs and is a fixed small integer in the L-BFGS method. When fewer than  $p$  curvature pairs are available, the BFGS update is applied using all currently available pairs.

Now, we provide a sufficient condition for Assumption 3 by the following proposition, which is a variant of existing work [17, Theorem 5.5] [19, Lemma 1]. Its proof is deferred to section A.2.

**Proposition 2.** *Let  $\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$  be a fixed number of curvature pairs with  $s_{k_i} \neq 0$ . Assume that there exist constants  $0 < \lambda \leq \Lambda$  such that for all  $(s, y) \in \{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}$ ,*

$$y^\top s \geq \frac{1}{\Lambda} \|y\|^2, \quad (3.36)$$

$$y^\top s \geq \lambda \|s\|^2. \quad (3.37)$$

Define the condition number  $\kappa := \Lambda/\lambda$ . Then there exist  $0 < m < M$  such that

$$mI \preceq \text{BFGS}(\{(s_{k_i}, y_{k_i})\}_{i=0}^{p-1}) \preceq MI, \quad (3.38)$$

where

$$M := \gamma + p\Lambda, \quad m := \left( (1 + \sqrt{\kappa})^{2p} \left( \gamma + \frac{1}{\lambda(2\sqrt{\kappa} + \kappa)} \right) \right)^{-1}.$$

Proposition 2 shows that, whenever eqs. (3.36) and (3.37) hold for the stored pairs, the resulting BFGS matrix satisfies Assumption 3 as expressed in eq. (3.38).

Based on Proposition 2, we basically adopt the following rule for forming  $B_k$ :

$$B_k = \text{BFGS}(\{(s_{k_i}, \bar{y}_{k_i})\}), \quad (3.39)$$

where

$$\bar{y}_{k_i} := \theta_i^{\text{damp}} y_{k_i} + (1 - \theta_i^{\text{damp}}) B_{k-1} s_{k_i}$$

with  $\theta_i^{\text{damp}} \in [0, 1]$ . This parameter is determined using a damped BFGS update [32, 41] and the indices  $\{k_i\}_{i=0}^{p-1}$  are selected so that the modified pairs  $(s_{k_i}, \bar{y}_{k_i})$  satisfy eqs. (3.36) and (3.37) for some constants  $\lambda, \Lambda$ . The damped BFGS is a classical technique to ensure  $\bar{y}_{k_i}^\top s_{k_i} > 0$ , and we adopted its simple limited-memory variant [32].

Since we do not employ Wolfe conditions in the line search, this damping technique plays a central role in ensuring the positive definiteness of  $B_k$ . The selection of curvature pairs shares the same spirit as the cautious update [29, 33, 34], which only uses pairs satisfying certain curvature conditions. By Proposition 2,  $B_k$  satisfies the uniform spectral bound in Assumption 3. This confirms that the practical construction of  $B_k$  is consistent with the theoretical requirements.

**Remarks for Practical Implementation:** Having established that the proposed construction of  $B_k$  satisfies Assumption 3, we now present several remarks on its practical implementation. We begin by describing how to compute the search direction

$$d_k = -(B_k + \mu_k I)^{-1} g_k \quad (3.40)$$

with  $B_k$  in eq. (3.39). When  $\mu_k = 0$ , the direction can be computed efficiently by the standard L-BFGS two-loop recursion [31, 38]. For the case  $\mu_k > 0$ , regularized limited-memory schemes may in principle be employed [27]. Alternatively, one may adopt the simple and robust approximation mentioned in the same reference. Specifically, we approximate  $B_k$  in eq. (3.39) by

$$B_k \approx \text{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + \mu_k s_{k_i}\}_{i=0}^{p-1}) - \mu_k I, \quad (3.41)$$

so that  $B_k + \mu_k I \approx \text{BFGS}(\{s_{k_i}, \bar{y}_{k_i} + \mu_k s_{k_i}\}_{i=0}^{p-1})$ . Its inverse can then be computed using the standard two-loop recursion. Although this approximation is inexact, it has been reported to yield nearly identical practical performance while remaining comparatively simple to implement [27]. We adopt this approximation in our numerical experiments.

Separately, there exist function-based modified secant conditions [2, 24, 54, 58] that adjust the secant equation using available function values and gradients when those quantities are deemed reliable. Such adjustments aim to improve practical convergence behavior. We partially adopt these function-based modifications following prior work [55, 59]. The suffix -MS in section 3.5 indicates the use of this technique. Because these modifications are well established, we omit low-level implementation details here. Further specifics are available in the cited references and in our open-source implementation\*<sup>1</sup>.

### 3.4.2 Step Size $\alpha_k$

We next discuss the adjustment of the step size  $\alpha_k$  in line 4 of Algorithm 2.

As indicated in the pseudo-code, we choose  $\alpha_k$  using interpolation of the objective function. Moré–Thuente line searches [35] are widely used in quasi-Newton methods [49], and they typically employ cubic or quadratic interpolation to select trial step sizes. In our implementation, we use an essentially identical procedure with a slight modification, namely, clipping the trial step size to the interval  $[\beta_{\min} \alpha_k, \beta_{\max} \alpha_k]$ . For all numerical experiments, we set the Armijo parameter to  $c = 10^{-4}$  and choose the interpolation bounds as  $\beta_{\min} = 1/16$  and  $\beta_{\max} = 15/16$ .

Furthermore, when  $\mu_k > 0$  and  $\alpha_k = 1$ , we introduce a one-time adjustment of the trial step size using gradient information. Let  $d_k$  be the search direction and  $g_k$  and  $g_{\text{try}}$  the gradients at the current and trial points. If the directional derivative along  $d_k$  changes sign,  $d_k^\top g_k < 0$  and  $d_k^\top g_{\text{try}} > 0$ , and the trial gradient is sufficiently aligned with  $d_k$ , namely  $d_k^\top g_{\text{try}} > 0.5\|d_k\|\|g_{\text{try}}\|$ , we rescale  $\alpha_k$  once by a secant-like factor,

$$\alpha_k \leftarrow \text{clip}\left(\alpha_k \frac{-d_k^\top g_k}{d_k^\top g_{\text{try}} - d_k^\top g_k}, \beta_{\min} \alpha_k, \beta_{\max} \alpha_k\right),$$

where  $\text{clip}(x, a, b) := \min(\max(x, a), b)$  is a clipping function. When objective values are unreliable and backtracking and interpolation do not occur, this correction effectively refines the step size. Since this adjustment is at most once per iteration, the theoretical results in section 3.3 remain valid.

### 3.4.3 Regularization Parameter $\mu_k$

We finally discuss the adjustment of the regularization parameter  $\mu_k$  in line 8 of Algorithm 1.

In our implementation, we choose the regularization parameter as

$$\mu_k = \text{clip}\left(\frac{1}{10}\|g_k\|, \frac{1}{100}G_k, G_k\right) \quad \text{where} \quad G_k := \sqrt{\varsigma + \sum_{j \in K, j \leq k} \|g_j\|^2},$$

which is consistent with the general form (eq. (3.6)) with an adaptive factor  $\theta_k \in [10^{-2}, 1]$ . In practice, we may set  $\varsigma = 0$  as long as the initial gradient is nonzero since  $\varsigma$  is mainly introduced for avoiding zero division, but to make it consistent with the theory, we set  $\varsigma = 10^{-10}$  in all experiments.

When  $B_k$  is a good approximation of the Hessian, the regularization parameter  $\mu_k$  should be small for fast convergence. Since the quantity  $G_k$  is strictly increasing with  $k$ , it is natural in practice to decrease the effective regularization when sufficient progress in the objective function is achieved. When the difference between the two sides of eq. (3.5) is larger than 1, we restart the accumulation by resetting  $K^+$  to the empty set and  $G_k$  to  $\varsigma$ . As long as the number of such restarts is finite, the overall convergence analysis in section 3.3 remains valid. Since these choices are heuristic, there may be room for further improvement. Especially, further increasing  $\mu_k$  stabilizes the method when the objective function values are highly noisy, and decreasing  $\mu_k$  more aggressively may accelerate convergence when the objective function is accurate.

## 3.5 Numerical Experiments

### 3.5.1 Methods

We compared the following optimization algorithms:

- (Ours): Our proposed regularized quasi-Newton method
  - We used Algorithms 1 and 2 with the details in section 3.4.
  - (Ours-MS) means the variant incorporating the modified secant condition described in section 3.4.1.
- (Line): Line search L-BFGS method
  - Our Python implementation ported from LBFGSpp [39].
  - The step size was determined using the Moré–Thuente line search, closely resembling SciPy’s implementation.
  - (Line-MS) means the same variant as Ours-MS.

- (Reg): Regularized L-BFGS method [27]
  - A quadratic regularization-based quasi-Newton method that does not rely on line search and is conceptually closer to trust-region methods.
  - We wrapped the function `regLBFGS` with `solveNonmonotone` from [27].
  - (Reg-Sec) means the incorporation of the approximated computation as in eq. (3.41). This is a wrapped function of `regLBFGSsec` from [27].
- (SciPy): SciPy’s L-BFGS-B method [49]
  - An established L-BFGS method implemented in C and called from Python.
  - We used the default parameters, with the exception that `ftol` was set to 0 to avoid premature termination.
- (NTQN): Noise-tolerant quasi-Newton method [45, 53]
  - A method designed to accommodate inexact gradient information, as mentioned in section 3.1.3.
  - We mainly used the default parameter settings, and slightly modified the stopping criteria to terminate at the desired gradient norm tolerance.

In the following, each method is referred to by the name given in parentheses. All these methods are L-BFGS-based, and we set the number of stored curvature pairs to 10, the default value in SciPy’s L-BFGS-B implementation.

All experiments were conducted on a Microsoft Windows 11 Home system (version 10.0, build 26100) equipped with a 13th Gen Intel® Core™ i7-1360P CPU, featuring 12 physical cores and 16 logical processors. All computations were performed using the CPU only.

### 3.5.2 Experimental Results

#### Test Problems and Settings

We evaluated all algorithms on unconstrained problems from the CUTEst benchmark collection [18], implemented in Python using the PyCUTEst interface [16]. All problems were initialized with the default starting points provided by the library. Following the previous work [27], we excluded problems such as the initial point is already stationary or contains NaN and INF in its function values or gradient. To simulate different numerical environments, we tested the following four settings.

- Artificially noised 64-bit floating-point precision (220 problems,  $\varepsilon_f = 10^{-2}$ )
- 64-bit floating-point precision (220 problems,  $\varepsilon_f = 2.22 \times 10^{-9}$ )
- 32-bit floating-point precision (220 problems,  $\varepsilon_f = 1.19 \times 10^{-2}$ )
- 16-bit floating-point precision (152 problems,  $\varepsilon_f = 9.77 \times 10^{-1}$ )

In the artificial noise setting, we added uniform random noise to both the objective function values and each component of the gradient vectors independently, i.e.,

$$\bar{f}(x) = f(x) + \text{unif}(-10^{-3}, 10^{-3}), \quad \nabla \bar{f}(x) = \nabla f(x) + \text{unif}(-10^{-3}, 10^{-3}) \mathbb{1},$$

where  $\text{unif}(a, b)$  denotes a uniform random variable in the interval  $[a, b]$  and  $\mathbb{1}$  is the vector of all ones. This setting is the same as the one in [45]. In the reduced precision settings, to simulate lower numerical precision, we first cast the input vector  $x$  to the target lower precision, then computed  $\bar{f}(x)$  and  $\nabla \bar{f}(x)$  using the lower-precision input. Although the internal computations of CUTEst remain in 64-bit precision, the casting of the input vector effectively simulated the impact of reduced precision on function and gradient evaluations.

Each setting uses a different error rate  $\varepsilon_f$  in the error model (eq. (3.1)), as summarized in Table 3.1. The table relates the machine epsilon (maximum relative rounding error) for each floating-point precision to the  $\varepsilon_f$  value used in our experiments. The relatively large  $\varepsilon_f$  values are chosen to account for the cumulative errors that may arise during optimization. We also set the

| explanation             | min abs value           | relative error                         | $\varepsilon_f$       |
|-------------------------|-------------------------|----------------------------------------|-----------------------|
| 64-bit double precision | $2.23 \times 10^{-308}$ | $2^{-52} \approx 2.22 \times 10^{-16}$ | $2.22 \times 10^{-9}$ |
| 32-bit single precision | $1.18 \times 10^{-38}$  | $2^{-23} \approx 1.19 \times 10^{-7}$  | $1.19 \times 10^{-3}$ |
| 16-bit half precision   | $6.10 \times 10^{-5}$   | $2^{-10} \approx 9.77 \times 10^{-4}$  | $9.77 \times 10^{-2}$ |

Table 3.1: The “min abs value” is the smallest positive normalized number, the “relative error” is the maximum relative rounding error, and  $\varepsilon_f$  is the parameter in (??) used in our experiments.

maximum number of iterations to  $k_{\max} = 15,000$  and timed out runs exceeding 10 minutes per problem.

#### Performance Profiles

We used performance profiles [14] to compare the algorithms. Let  $P$  denote the set of test problems and  $S$  the set of solvers. For each problem  $p \in P$  and solver  $s \in S$ , let  $n_{p,s} > 0$  denote the number of function and gradient evaluations required to reach a specified gradient norm tolerance  $\varepsilon_{\text{gtol}}$ . Following common implementations [49], we used the infinity norm for the gradient, though other norms may equally be employed. Note that  $n_{p,s}$  is not time in this experiment but the number of evaluations. When a solver fails to converge within the gradient tolerance or exceeds a maximum number of evaluations, we set  $n_{p,s} = +\infty$ . Then, the performance ratio  $r_{p,s}$  is defined as

$$r_{p,s} = \frac{n_{p,s}}{\min_{s' \in S} n_{p,s'}}.$$

For a given threshold  $\tau \geq 1$  and solver  $s$ , the performance profile plots the proportion of problems for which  $r_{p,s} \leq \tau$ . This means that the  $y$ -coordinate for solver  $s$  at  $x$ -coordinate  $\tau$  is given by

$$\rho_s(\tau) = \frac{1}{|P|} |\{p \in P \mid r_{p,s} \leq \tau\}|.$$

A curve that lies above others indicates better performance.

#### Results with Artificial Noise

We first present the result in the artificial noise setting described in section 3.5.2. Figure 3.1 presents the performance profile. In this regime, the proposed method shows superior robustness compared to other quasi-Newton methods. This result highlights the validity of our theoretical analysis in section 3.3 and demonstrates the effectiveness of our approach in handling noisy objective function values.

#### Results at Different Precision Levels

We next present performance profiles for each floating-point precision (64-bit, 32-bit, and 16-bit). We evaluated each method at three gradient norm tolerance levels, i.e.,  $\varepsilon_{\text{gtol}} \in \{10^{-1}, 10^{-3}, 10^{-5}\}$ . These tolerance levels represent different optimization regimes, from moderate accuracy to high precision requirements. Figure 3.2 shows the comprehensive performance profiles across all three precision levels. Our methods (Ours) and (Ours-MS) demonstrates competitive or superior performance compared to existing methods in the standard 64-bit setting. As numerical precision degrades from 64-bit to 32-bit and further to 16-bit, our regularization-based approach demonstrates improved relative performance. The proposed method maintains stability even when traditional line search methods encounter difficulties with reduced precision arithmetic.



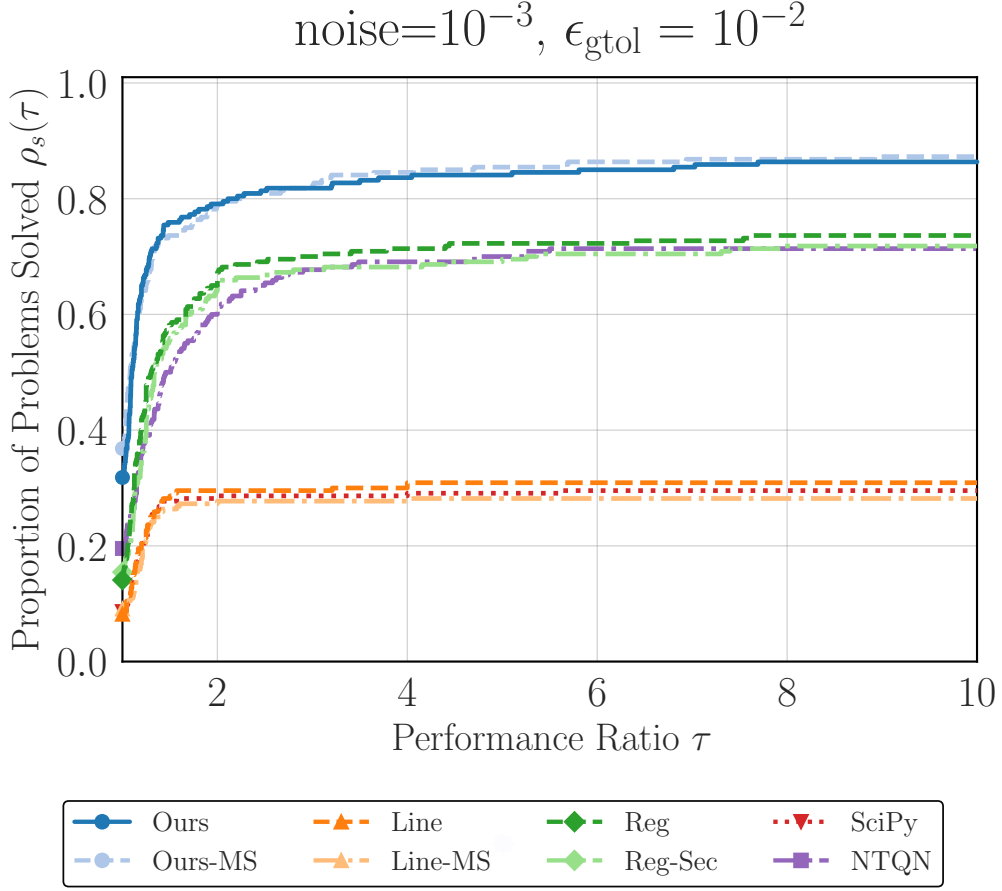


Fig. 3.1: Performance profile for artificial noise setting (level of  $10^{-3}$ ) and  $\epsilon_{\text{gtol}} = 10^{-2}$ .

### 3.5.3 Computational Time per Iteration

In this subsection, we present an experiment to compare the per-iteration computational cost of each method. Here, our purpose lies in the measurement of the total execution time, not the number of oracle calls. We conducted experiments on the ill-conditioned quadratic problem

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2} \sum_{i=1}^n ix_i^2$$

with  $n = 10,000$ . The initial point  $x_0$  was chosen as the all-ones vector. We used memory size  $m = 10$  and maximum iterations  $k_{\text{max}} = 100$ , and we did not set any stopping criteria other than reaching  $k_{\text{max}}$ . The simplicity of the optimization problem and the small value of  $k_{\text{max}}$  are well suited to the purpose of this experiment, as they ensure that the total runtime is dominated by algorithmic overhead rather than function or gradient computation. Timing data were collected after three warm-up runs, followed by 100 recorded trials.

In Figure 3.3, we report the execution time statistics, and we can observe that our proposed methods (Ours) and (Ours-MS) exhibit competitive performance compared to other methods. Combining with the results from section 3.5.2, our methods is competitive or superior in total oracle calls and per-iteration computational cost, indicating practical efficiency. As a side note, although the (SciPy) calls optimized C routines, it incurs associated dispatch overhead, produc-



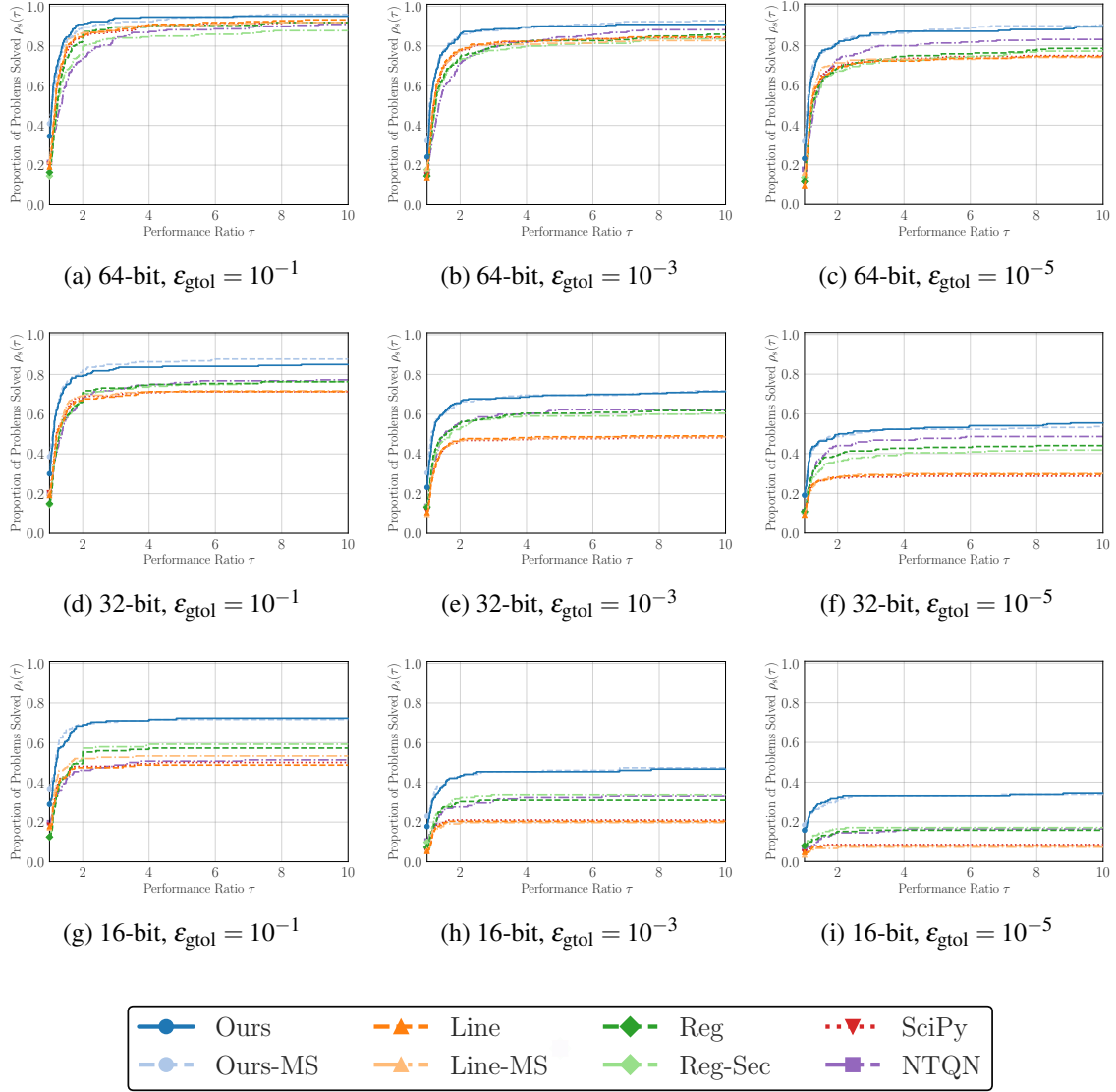


Fig. 3.2: Performance profiles across different floating-point precisions (64, 32, and 16-bit) and tolerance levels. The results demonstrate the robustness of the proposed method.

ing a larger mean time. One of the reasons of the (Reg) and (Reg-Sec) methods taking longer execution time is due to the implementation aspect of the original code.

In Table 3.2, we report the total number of oracle calls (calls of  $\bar{f}$  and  $\nabla \bar{f}$ ) and final objective values  $f(x_{k_{\max}})$ . The purpose of this table is to confirm that all methods achieved similar convergence behavior within the limited number of iterations. The comparable values in the table indicate that the comparison of execution times is fair and not affected by differences in convergence behavior.

### 3.6 Conclusion

In this work, we proposed a regularized quasi-Newton method for noisy nonconvex optimization and established its global convergence properties. We also demonstrated the method's effectiveness through numerical experiments on standard benchmark problems under various noise levels

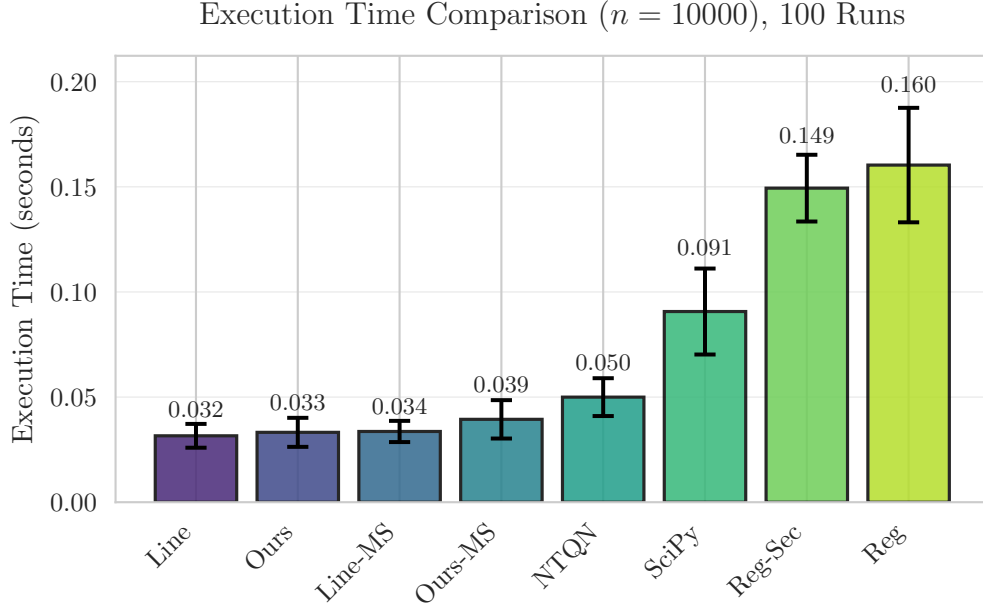


Fig. 3.3: Mean execution times for the experiment in section 3.5.3. This plot roughly indicates the extra computational overhead of each method. Error bars represent one standard deviation over 100 runs.

|                         | Line | Ours | Line-MS | Ours-MS | NTQN | SciPy | Reg-Sec | Reg  |
|-------------------------|------|------|---------|---------|------|-------|---------|------|
| # Calls                 | 212  | 202  | 212     | 202     | 219  | 212   | 206     | 206  |
| $\bar{f}(x_{k_{\max}})$ | 1.24 | 1.34 | 1.24    | 1.34    | 1.43 | 1.24  | 1.48    | 1.47 |

Table 3.2: The number of oracle calls and final observed objective values for the experiment in section 3.5.3.

and precision settings. The results indicate that our method is robust and effective, particularly in noisy or low-precision environments where traditional line search methods may struggle.

Future work includes the investigation of local convergence properties, the extension to constrained optimization, and the applications to practical problems in machine learning and scientific computing. Additionally, exploring adaptive strategies for selecting algorithmic parameters based on problem characteristics could further enhance the method's performance. An explicit analysis of the effect of inexact gradient evaluations would also provide valuable insight into the method's robustness and guide further enhancements.

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# Bibliography

- [1] Babaie-Kafaki S (2011) A modified BFGS algorithm based on a hybrid secant equation. *Science China Mathematics* 54(9):2019–2036, DOI 10.1007/s11425-011-4232-7
- [2] Babaie-Kafaki S, Ghanbari R, Mahdavi-Amiri N (2010) Two new conjugate gradient methods based on modified secant equations. *Journal of Computational and Applied Mathematics* 234(5):1374–1386, DOI 10.1016/j.cam.2010.01.052
- [3] Baillon JB, Haddad G (1977) Quelques propriétés des opérateurs angle-bornés et cycliquement monotones. *Israel Journal of Mathematics* 26(2):137–150, DOI 10.1007/BF03007664
- [4] Bauschke HH, Combettes PL (2009) The Baillon-Haddad Theorem Revisited. DOI 10.48550/arXiv.0906.0807, [0906.0807](#)
- [5] Bellavia S, Gurioli G, Morini B, Toint PL (2021) The Impact of Noise on Evaluation Complexity: The Deterministic Trust-Region Case. DOI 10.48550/arXiv.2104.02519, [2104.02519](#)
- [6] Berahas AS, Cao L, Choromanski K, Scheinberg K (2021) A Theoretical and Empirical Comparison of Gradient Approximations in Derivative-Free Optimization. DOI 10.48550/arXiv.1905.01332, [1905.01332](#)
- [7] Berahas AS, Curtis FE, Zhou B (2022) Limited-memory BFGS with displacement aggregation. *Mathematical Programming* 194(1):121–157, DOI 10.1007/s10107-021-01621-6
- [8] Byrd RH, Nocedal J (1989) A Tool for the Analysis of Quasi-Newton Methods with Application to Unconstrained Minimization. *SIAM Journal on Numerical Analysis* 26(3):727–739, DOI 10.1137/0726042
- [9] Caffisch RE (1998) Monte Carlo and quasi-Monte Carlo methods. *Acta Numerica* 7:1–49, DOI 10.1017/S0962492900002804
- [10] Carter RG (1993) Numerical Experience with a Class of Algorithms for Nonlinear Optimization Using Inexact Function and Gradient Information. *SIAM Journal on Scientific Computing* 14(2):368–388, DOI 10.1137/0914023
- [11] Clancy RJ, Menickelly M, Hückelheim J, Hovland P, Nalluri P, Gjini R (2022) TROPHY: Trust Region Optimization Using a Precision Hierarchy. In: *Computational Science – ICCS 2022*, Springer, Cham, pp 445–459, DOI 10.1007/978-3-031-08751-6\_32
- [12] Corless RM, Gonnet GH, Hare DEG, Jeffrey DJ, Knuth DE (1996) On the LambertW function. *Advances in Computational Mathematics* 5(1):329–359, DOI 10.1007/BF02124750
- [13] Doikov N, Mishchenko K, Nesterov Y (2024) Super-Universal Regularized Newton Method. *SIAM Journal on Optimization* 34(1):27–56, DOI 10.1137/22M1519444
- [14] Dolan ED, Moré JJ (2002) Benchmarking optimization software with performance profiles. *Mathematical Programming* 91(2):201–213, DOI 10.1007/s101070100263
- [15] Duchi J, Hazan E, Singer Y (2011) Adaptive Subgradient Methods for Online Learning and Stochastic Optimization. *Journal of Machine Learning Research* 12(61):2121–2159
- [16] Fowkes J, Roberts L, Búrmen Á (2022) PyCUTEst: An open source Python package of optimization test problems. *Journal of Open Source Software* 7(78):4377, DOI 10.21105/joss.04377
- [17] Gao W, Goldfarb D (2018) Quasi-Newton Methods: Superlinear Convergence Without Line Searches for Self-Concordant Functions. DOI 10.48550/arXiv.1612.06965, [1612.06965](#)

- [18] Gould NI, Orban D, Toint PL (2015) CUTEst: A Constrained and Unconstrained Testing Environment with safe threads for mathematical optimization. *Comput Optim Appl* 60(3):545–557, DOI 10.1007/s10589-014-9687-3
- [19] Gower RM, Goldfarb D, Richtárik P (2016) Stochastic Block BFGS: Squeezing More Curvature out of Data. DOI 10.48550/arXiv.1603.09649, [1603.09649](#)
- [20] Grapiglia GN, Stella GFD (2022) An adaptive trust-region method without function evaluations. *Computational Optimization and Applications* 82(1):31–60, DOI 10.1007/s10589-022-00356-0
- [21] Gratton S, Toint PL (2020) A note on solving nonlinear optimization problems in variable precision. *Computational Optimization and Applications* 76(3):917–933, DOI 10.1007/s10589-020-00190-2
- [22] Gratton S, Toint PL (2023) OFFO minimization algorithms for second-order optimality and their complexity. *Computational Optimization and Applications* 84(2):573–607, DOI 10.1007/s10589-022-00435-2
- [23] Gratton S, Jerad S, Toint PhL (2024) Complexity of a class of first-order objective-function-free optimization algorithms. *Optimization Methods and Software* 0(0):1–31, DOI 10.1080/10556788.2023.2296431
- [24] Hassan BA, Moghrabi IAR (2023) A modified secant equation quasi-Newton method for unconstrained optimization. *Journal of Applied Mathematics and Computing* 69(1):451–464, DOI 10.1007/s12190-022-01750-x
- [25] Higham NJ (2002) *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, DOI 10.1137/1.9780898718027
- [26] Izycheva A, Darulova E (2017) On sound relative error bounds for floating-point arithmetic. In: *Proceedings of the 17th Conference on Formal Methods in Computer-Aided Design, FMCAD Inc, Austin, Texas*
- [27] Kanzow C, Steck D (2023) Regularization of limited memory quasi-Newton methods for large-scale nonconvex minimization. *Mathematical Programming Computation* 15(3):417–444, DOI 10.1007/s12532-023-00238-4
- [28] Lee M, Sun Y (2025) Advancing Multi-Secant Quasi-Newton Methods for General Convex Functions. DOI 10.48550/arXiv.2504.07330, [2504.07330](#)
- [29] Li DH, Fukushima M (2001) On the Global Convergence of the BFGS Method for Nonconvex Unconstrained Optimization Problems. *SIAM Journal on Optimization* 11(4):1054–1064, DOI 10.1137/S1052623499354242
- [30] Li YJ, Li DH (2009) Truncated regularized Newton method for convex minimizations. *Computational Optimization and Applications* 43(1):119–131, DOI 10.1007/s10589-007-9128-7
- [31] Liu DC, Nocedal J (1989) On the limited memory BFGS method for large scale optimization. *Mathematical Programming* 45(1):503–528, DOI 10.1007/BF01589116
- [32] Lotfi S, de Ruisselet TB, Orban D, Lodi A (2020) Stochastic Damped L-BFGS with Controlled Norm of the Hessian Approximation. DOI 10.13140/RG.2.2.27851.41765/1, [2012.05783](#)
- [33] Mannel F (2025) A Globalization of L-BFGS and the Barzilai–Borwein Method for Nonconvex Unconstrained Optimization. *Journal of Optimization Theory and Applications* 204(3):1–34, DOI 10.1007/s10957-024-02565-5
- [34] Mannel F, Om Aggrawal H, Modersitzki J (2024) A structured L-BFGS method and its application to inverse problems. *Inverse Problems* 40(4):045022, DOI 10.1088/1361-6420/ad2c31
- [35] Moré JJ, Thuente DJ (1994) Line search algorithms with guaranteed sufficient decrease. *ACM Trans Math Softw* 20(3):286–307, DOI 10.1145/192115.192132
- [36] Moré JJ, Wild SM (2011) Estimating Computational Noise. *SIAM Journal on Scientific Computing* 33(3):1292–1314, DOI 10.1137/100786125
- [37] Nesterov Y, Polyak B (2006) Cubic regularization of Newton method and its global perfor-

- mance. *Mathematical Programming* 108(1):177–205, DOI 10.1007/s10107-006-0706-8
- [38] Nocedal J, Wright SJ (1999) *Numerical Optimization*. Springer, DOI 10.1007/978-0-387-40065-5
- [39] Okazaki N (2007) libLBFGS: A library of limited-memory broyden-fletcher-goldfarb-shanno (l-BFGS)
- [40] Paquette C, Scheinberg K (2020) A Stochastic Line Search Method with Expected Complexity Analysis. *SIAM Journal on Optimization* 30(1):349–376, DOI 10.1137/18M1216250
- [41] Powell MJ (1978) Algorithms for nonlinear constraints that use lagrangian functions. *Math Program* 14(1):224–248, DOI 10.1007/BF01588967
- [42] Ranganath A, Singhal M, Marcia R (2025) Symmetric Rank-One Quasi-Newton Methods for Deep Learning Using Cubic Regularization. DOI 10.48550/arXiv.2502.12298, [2502.12298](#)
- [43] Ruan H, Yang W (2024) Adaptive Regularized Quasi-Newton Method Using Inexact First-Order Information. *Journal of Computational Mathematics* 42(6):1656–1687, DOI 10.4208/jcm.2306-m2022-0279
- [44] Sheng Z, Yuan G, Cui Z (2018) A new adaptive trust region algorithm for optimization problems. *Acta Mathematica Scientia* 38(2):479–496, DOI 10.1016/S0252-9602(18)30762-8
- [45] Shi HJM, Xie Y, Byrd R, Nocedal J (2022) A Noise-Tolerant Quasi-Newton Algorithm for Unconstrained Optimization. *SIAM Journal on Optimization* 32(1):29–55, DOI 10.1137/20M1373190
- [46] Sun S, Nocedal J (2023) A trust region method for noisy unconstrained optimization. *Mathematical Programming* 202(1):445–472, DOI 10.1007/s10107-023-01941-9
- [47] Ueda K, Yamashita N (2010) Convergence Properties of the Regularized Newton Method for the Unconstrained Nonconvex Optimization. *Applied Mathematics and Optimization* 62(1):27–46, DOI 10.1007/s00245-009-9094-9
- [48] Ueda K, Yamashita N (2014) A regularized Newton method without line search for unconstrained optimization. *Computational Optimization and Applications* 59(1):321–351, DOI 10.1007/s10589-014-9656-x
- [49] Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, Burovski E, Peterson P, Weckesser W, Bright J, van der Walt SJ, Brett M, Wilson J, Millman KJ, Mayorov N, Nelson ARJ, Jones E, Kern R, Larson E, Carey CJ, Polat I, Feng Y, Moore EW, VanderPlas J, Laxalde D, Perktold J, Cimrman R, Henriksen I, Quintero EA, Harris CR, Archibald AM, Ribeiro AH, Pedregosa F, van Mulbregt P, SciPy 10 Contributors (2020) SciPy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods* 17:261–272, DOI 10.1038/s41592-019-0686-2
- [50] Wang S, Fadili J, Ochs P (2025) Convergence rates of regularized quasi-Newton methods without strong convexity. DOI 10.48550/arXiv.2506.00521, [2506.00521](#)
- [51] Ward R, Wu X, Bottou L (2020) AdaGrad stepsizes: Sharp convergence over nonconvex landscapes. *J Mach Learn Res* 21(1):219:9047–219:9076
- [52] Wei Z, Li G, Qi L (2006) New quasi-Newton methods for unconstrained optimization problems. *Applied Mathematics and Computation* 175(2):1156–1188, DOI 10.1016/j.amc.2005.08.027
- [53] Xie Y, Byrd RH, Nocedal J (2020) Analysis of the BFGS method with errors. *SIAM Journal on Optimization* 30(1):182–209
- [54] Yabe H, Ogasawara H, Yoshino M (2007) Local and superlinear convergence of quasi-Newton methods based on modified secant conditions. *Journal of Computational and Applied Mathematics* 205(1):617–632, DOI 10.1016/j.cam.2006.05.018
- [55] Yuan G, Wei Z (2010) Convergence analysis of a modified BFGS method on convex minimizations. *Computational Optimization and Applications* 47(2):237–255, DOI 10.1007/s10589-008-9219-0
- [56] Yuan YX (1991) A Modified BFGS Algorithm for Unconstrained Optimization. *IMA Journal of Numerical Analysis* 11(3):325–332, DOI 10.1093/imanum/11.3.325

- [57] Zhang H, Ni Q (2018) A new regularized quasi-Newton method for unconstrained optimization. *Optimization Letters* 12(7):1639–1658, DOI 10.1007/s11590-018-1236-z
- [58] Zhang J, Xu C (2001) Properties and numerical performance of quasi-Newton methods with modified quasi-Newton equations. *Journal of Computational and Applied Mathematics* 137(2):269–278, DOI 10.1016/S0377-0427(00)00713-5
- [59] Zhang JZ, Deng NY, Chen LH (1999) New Quasi-Newton Equation and Related Methods for Unconstrained Optimization. *Journal of Optimization Theory and Applications* 102(1):147–167, DOI 10.1023/A:1021898630001

## A

## Appendix

## A.1 Proof of Lemma 6

The proof of Lemma 6 is similar to existing results [51, Lemma 3.2], [23, Lemma 3.1].

*Proof.* Recall that eq. (3.6) defines  $\mu_k$  as  $\theta_k \sqrt{\varsigma + \sum_{j \in K^+, j \leq k} \|g_j\|^2}$ , and line 8 of Algorithm 1 asserts  $\theta_k \in [\theta_{\min}, \theta_{\max}]$ . Let  $k_i \in K^+$  be the  $i$ -th element of  $K^+$ . For any  $0 < x < y$ ,  $x/y \leq -\ln(1 - x/y) = \ln y - \ln(y - x)$  holds. Thus, we have

$$\begin{aligned} \frac{\|g_{k_i}\|^2}{\mu_{k_i}^2} &\leq \frac{1}{\theta_{\min}^2} \frac{\|g_{k_i}\|^2}{\varsigma + \sum_{\ell=0}^i \|g_{k_\ell}\|^2} && \text{(definition)} \\ &\leq \frac{1}{\theta_{\min}^2} \left( \ln \left( \varsigma + \sum_{\ell=0}^i \|g_{k_\ell}\|^2 \right) - \ln \left( \varsigma + \sum_{\ell=0}^{i-1} \|g_{k_\ell}\|^2 \right) \right). && (x/y \leq \ln y - \ln(y - x)) \end{aligned}$$

Similarly,  $x/\sqrt{y} \geq x/(\sqrt{y} + \sqrt{y - x}) = \sqrt{y} - \sqrt{y - x}$  holds. Thus, we have

$$\begin{aligned} \frac{\|g_{k_i}\|^2}{\mu_{k_i}} &\geq \frac{1}{\theta_{\max}} \frac{\|g_{k_i}\|^2}{\sqrt{\varsigma + \sum_{\ell=0}^i \|g_{k_\ell}\|^2}} && \text{(definition)} \\ &\geq \frac{1}{\theta_{\max}} \left( \sqrt{\varsigma + \sum_{\ell=0}^i \|g_{k_\ell}\|^2} - \sqrt{\varsigma + \sum_{\ell=0}^{i-1} \|g_{k_\ell}\|^2} \right). && (x/\sqrt{y} \geq \sqrt{y} - \sqrt{y - x}) \end{aligned}$$

Summing up these inequalities over  $0 \leq i < |K^+|$  yields the claim.  $\square$

## A.2 Proof of Proposition 2

The proof of Proposition 2 is similar to existing results [17, Theorem 5.5] [19, Lemma 1].

*Proof.* We consider standard BFGS updates with a single curvature pair  $(s, y)$ . From eqs. (3.36) and (3.37), we have

$$\left\| \frac{yy^\top}{y^\top s} \right\| = \frac{y^\top y}{y^\top s} \leq \Lambda, \quad \left\| \frac{ys^\top}{y^\top s} \right\| \leq \frac{\|y\| \|s\|}{\sqrt{\frac{1}{\Lambda} \|y\|^2} \sqrt{\lambda \|s\|^2}} = \sqrt{\kappa}, \quad \left\| \frac{ss^\top}{y^\top s} \right\| = \frac{s^\top s}{y^\top s} \leq \frac{1}{\lambda}. \quad (\text{A.1})$$



Let  $B, H$  denote the positive definite Hessian approximation before the update and its inverse, and  $B_+, H_+$  the updated matrices. The BFGS update rule is

$$B_+ = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}, \quad H_+ = \left(I - \frac{sy^\top}{y^\top s}\right) H \left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}.$$

Since  $B$  is symmetric positive definite matrix and  $s \neq 0$ , we have

$$\|B_+\| \leq \|B\| + \left\| \frac{yy^\top}{y^\top s} \right\| \leq \|B\| + \Lambda, \quad (\text{A.2})$$

$$\begin{aligned} \|H_+\| &= \left\| \left(I - \frac{ys^\top}{y^\top s}\right)^\top H \left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s} \right\| \\ &\leq \left\| I - \frac{ys^\top}{y^\top s} \right\|^2 \|H\| + \left\| \frac{ss^\top}{y^\top s} \right\| \\ &\leq \left(1 + \left\| \frac{ys^\top}{y^\top s} \right\|\right)^2 \|H\| + \left\| \frac{ss^\top}{y^\top s} \right\| \\ &\leq (1 + \sqrt{\kappa})^2 \|H\| + \frac{1}{\lambda}. \end{aligned} \quad (\text{eq. (A.1)}) \quad (\text{A.3})$$

Applying eqs. (A.2) and (A.3) recursively over  $p$  updates and using the initial matrix  $\gamma I$ , we obtain

$$\begin{aligned} \left\| \text{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}) \right\| &\leq \gamma + p\Lambda, \\ \left\| (\text{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1}))^{-1} \right\| &\leq (1 + \sqrt{\kappa})^{2p} \gamma + \frac{1}{\lambda} \sum_{i=0}^{p-1} (1 + \sqrt{\kappa})^{2i} \\ &= (1 + \sqrt{\kappa})^{2p} \gamma + \frac{1}{\lambda} \frac{(1 + \sqrt{\kappa})^{2p} - 1}{(1 + \sqrt{\kappa})^2 - 1} \\ &\leq (1 + \sqrt{\kappa})^{2p} \left( \gamma + \frac{1}{\lambda(2\sqrt{\kappa} + \kappa)} \right). \end{aligned}$$

Therefore,  $\text{BFGS}(\{s_{k_i}, y_{k_i}\}_{i=0}^{p-1})$  satisfies the stated bounds in eq. (3.38).  $\square$

The conditions eqs. (3.36) and (3.37) are standard in the analysis of quasi-Newton methods [8, Theorem 2.1]. If  $f$  is  $L$ -smooth (Assumption 2) and convex, then by the Baillon–Haddad theorem [3, 4] its gradient is  $1/L$ -cocoercive:

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2 \quad (\text{A.4})$$

for all  $x, y \in \mathbb{R}^n$ . Applying this inequality with  $x = x_{k+1}$  and  $y = x_k$  yields eq. (3.36) with  $\Lambda = L$ .